MS SOLAR 3
52 MW PV PLANT
Sumrall Mississippi

Plant Control & Monitoring

USER MANUAL

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1 SCOPE

This document is the User Manual for the Monitoring and Control System that ITABB Power Grids Division is supplying for the MS Solar 3 Photo Voltaic Plant, located in Sumrall (Mississippi).
It describes the MicroSCADA configuration and utilization and the HMI interface with the Plant Controller for the management of Active and Reactive power.
This document is devoted to all users, gives an overview on system architecture, functions and configuration. Moreover it gives to plant operator helps to navigate in the system, and helps engineers and administrators, on system maintenance and improvement.

Please note that this manual is devoted to Sumrall project, all the standard documentation on MicroSCADA, and AC500 PLC is available on the standard documents.

1.1 Reference Documents

- OPP-17-830386-TP-00_Rev0 (Technical Proposal)
- 1MMC-10058-AXX-S001 (Functional Design Specification)

Standard Documents:

- 2VAA001150-204 MicroSCADA User Guide
- 2VAA001148-204 MicroSCADA Configuration Guide
- 2VAA000830-204 MicroSCADA Database Reference Guide
- 2VAA000724-204 MicroSCADA History Reference Guide
- 2VAA001149-204 MicroSCADA Display Builder User Guide
- 2VAA001151-204 MicroSCADA History User Guide
- 2VAA000725-204 MicroSCADA Scanner Reference Guide

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1.2 Acronyms

AC	Alternating Current
AI	Analog Input
AO	Analog Output
CN	Combiner box
DC	Direct Current
DI	Digital Input (Boolean Input)
DO	Digital Output (Boolean Output)
DNO	Distribution Network Operator
eBoP	electrical Balance of Plant
BOS	Balance of System
FAT	Factory Acceptance Test
FCD	Functional Control Document
HMI	Human - Machine Interface
HV	High Voltage
HW	Hardware
IO	Input Output
LV	Low Voltage
JB	Junction Box
KPI	Key Performance Indicator
MV	Medium Voltage
PLC	Programmable Logic Controller
POA	Plane of Array
POI	Point of Interconnection
PR	Performance Ratio
PV	Photo Voltaic
RRL	Ramp Rate Limiter
SW	Software
THV	Transformer MV/HV
TML	Technical Minimum Level
TMV	Transformer LV/MV
TSO	Transmission System Operator

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2 DESCRIPTION

The Plant control and monitoring System provide to the operator the HMI for monitoring and control of a photovoltaic power plant.

By means of "***ABB MicroSCADA***" it provides:

- Graphical pages for plant data monitoring.
- Graphical interface for interaction with Plant Devices.
- Alarms management.
- Reports management.

By means of "***ABB AC500***" it provides:

- Active power limitation at plant Point of Interface.
- Voltage Control at plant Point of Interface.
- Reactive power control at plant Point of Interface.
- Power Factor control at plant Point of Interface.

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2.1 Architecture

The Figure 2-1 shows the system architecture, for the Plant Control and Monitoring system, showing the main components and an overall of the data flow.

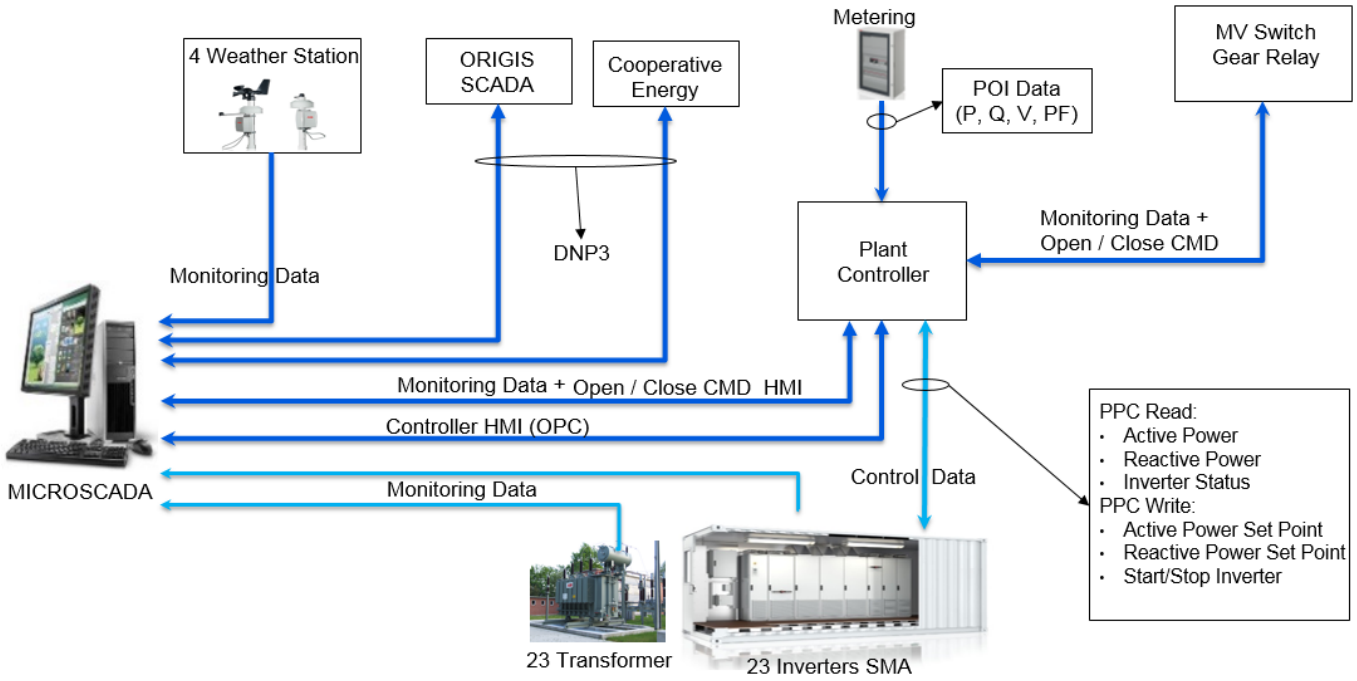

Figure 2-1 ***Plant Monitoring & Control Architecture***

Therefore the whole plant control system consists of the following HW and SW:

- Num.1 Industrial server mounted inside a cabinet.
- Operating system Server windows 2008 r2 sp2 64Bit.
- Office Excel license for DB management and reports.
- MicroSCADA software license with:
  - 7000 real time tag
  - History data logging
  - Calculation engine
  - 8 Modbus TCP (Master) and 2 DNP3 (Slave) protocols
  - 2 remote client.
- Num.1 Plant Controller Hardware based on ABB AC500 PLC:
  - Num.1 CPU PM590-ETH AC500, 2MB ram/flash-eprom, Integrated Ethernet, web-server, Integrated Serial communication.

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- Num.1 TB521-ETH base board, 1 slot coprocessor, 1 Field Bus Port, 2 RS232/485 Ports (ASCII, Modbus-RTU, CS31), 1 Ethernet RJ45 port.
- Num.2 CM597-ETH Communication Module Ethernet.
- Num.1 DI561 digital IO module (max.8 Digital Input).

- Num.1 Network Switch.

The control functions, for the power generation control will be realized on ABB AC500 PLC, while the monitoring, and reporting functions are in charge of the MicroSCADA, hosted on the industrial server.

MicroSCADA Pro SYS600 Control System is a modular and scalable automation product.
It is based on a real-time database that reads, via the proper driver interfaces, process data from connected devices. The process data are stored, together with status and time stamp, and made available for clients (local or remote) to read and visualize it.
The real time database also generates alarm messages, creates audible alarms, can do calculations and many more features.
MicroSCADA provides operators with detailed well-arranged process overview displays. Direct access navigation leads to information-rich control faceplates, superior trending capabilities, extended alarm management, and various reports.
The acquired data can be directly used, or they can be used to create new calculated data, that can be managed (as the acquired ones) for display presentation, alarm management and reporting.
Almost all the data are acquired from the field using the Modbus TCP/IP - RTU protocol. The only data acquired through wires are the ones of the transformer.
The DI module connected to the AC500 PLC, is intended for the data acquisition from cabinet auxiliaries (for instance the high temperature sensor, the power supply alarm, the door opened switch etc.).

From functional point of view the plant is composed by the following blocks:

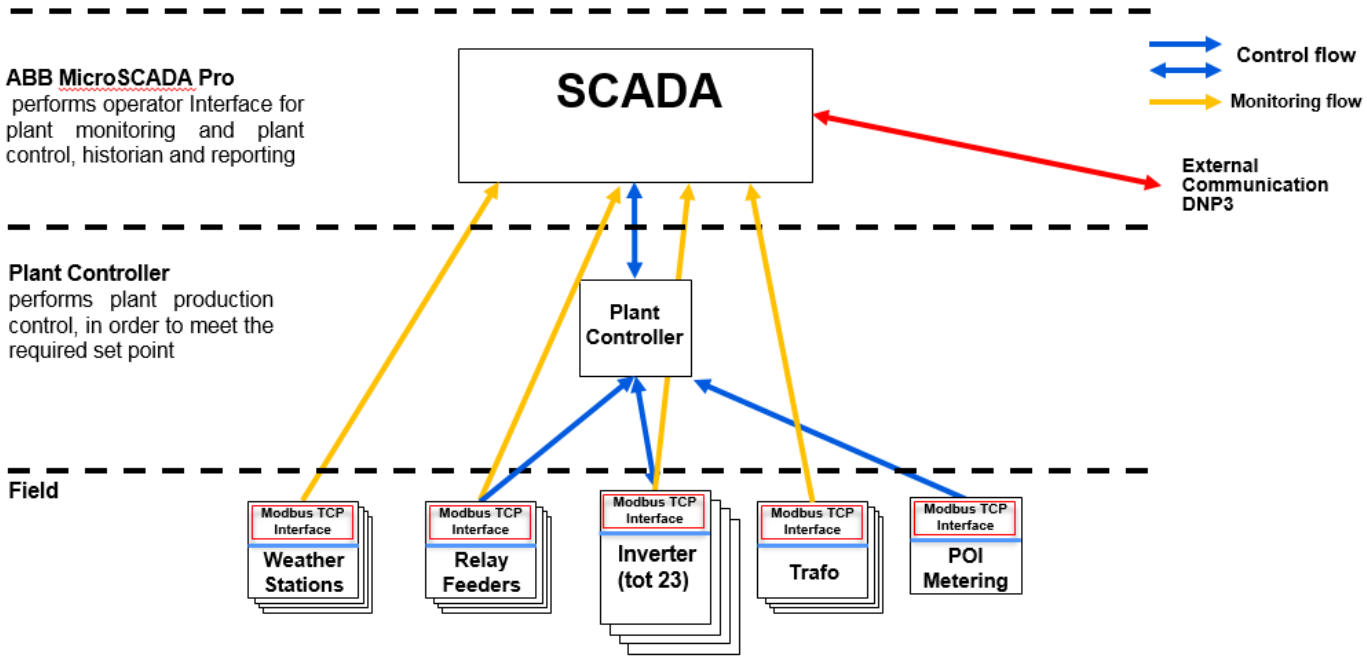

Figure 2-2 Functional Blocks

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- **Field** all inputs from external devices. They can be grouped in
  - Input from MV station Electrical Devices.
  - Input from MV station Electrical Measurements.
  - Input from Inverters.
  - Input from transformers.
  - Input from Plant Controller.
  - Input from Weather Stations.

- **Plant Controller** performing control of plant production compliant with Grid Manager Set-Point (Central Dispatch Center) or Utility Operator Set-Point.

- **MicroSCADA** performing Operator Interface, Monitoring, Calculation, Historian and Reporting.

Data flows from/to the blocks can be mainly divided in two groups:

- **Control Flows:** they have to be fast and they are bidirectional, Field and Plant Controller are directly connected. MicroSCADA acts as HMI for Plant Controller. Operator can manage set-points, control selection from mimics;

- **Monitoring Flows:** they are from Field to MicroSCADA real-time database.

The SCADA subsystems perform several actions:

- **Acquisition from Field or Plant Controller:** input data for MicroSCADA are recorded in real time database.

- **Calculation:** calculation engine cyclically executes calculation routine and records output data in the real time data base.

- **Historical archiving:** selected data from real time database are sent to data object for archiving.

- **Reporting:** report scheduler cyclically reads tags from data object following measurement report and exports results in txt or csv files.

- **Event/Alarm archiving:** alarm and event are recorded.

- **MicroSCADA HMI:** HMI uses mimics to show data from real-time database, trends to follow time variable behaviors using data from data object.
  Human Machine Interface is organized as a workplace where user can follow plant behavior and operate on it. Different user profiles enable different workplace functions.
  From HMI user can open event/alarm viewer to analyze the process.

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3 MICROSCADA START AND STOP

MicroSCADA service can be automatically activated after the system boot or activated in manual mode from Control Panel. You can select the 'Service Startup Type under Configuration menu:

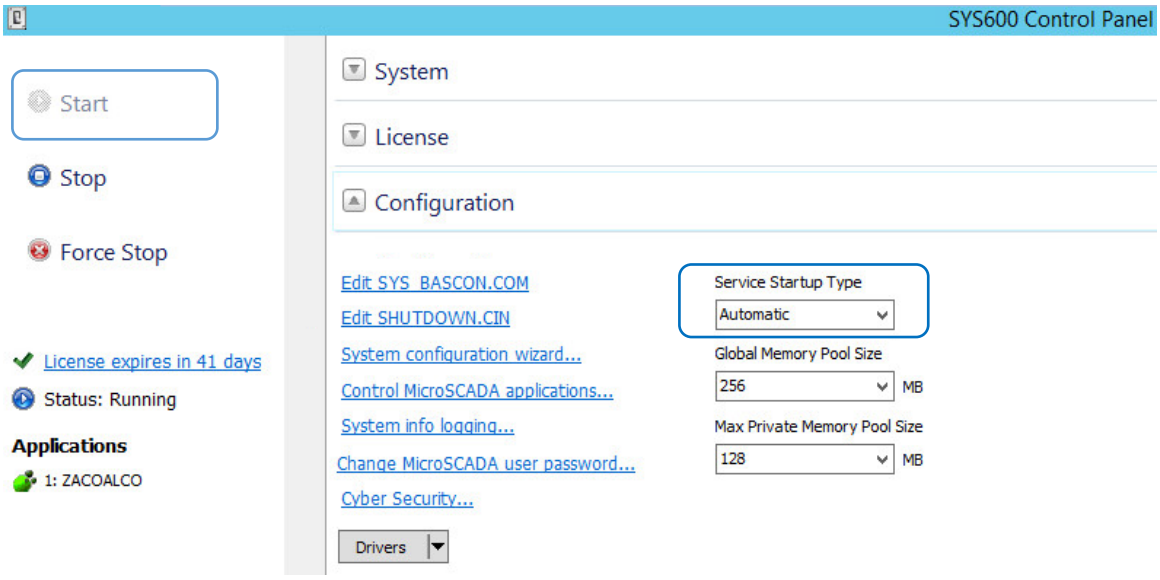

Figure 3-1 SYS600 Control Panel

In order to activate the user interface you have to start the MicroSCADA Monitor PRO, that is the user interface to MicroSCADA and Plant Controller.

To start MicroSCADA Monitor PRO, you can click on the shortcut on desktop:

On the desktop you will find also the shortcut for Control Panel:

At start-up MicroSCADA Monitor PRO opens a workplace where user can see the Home page:

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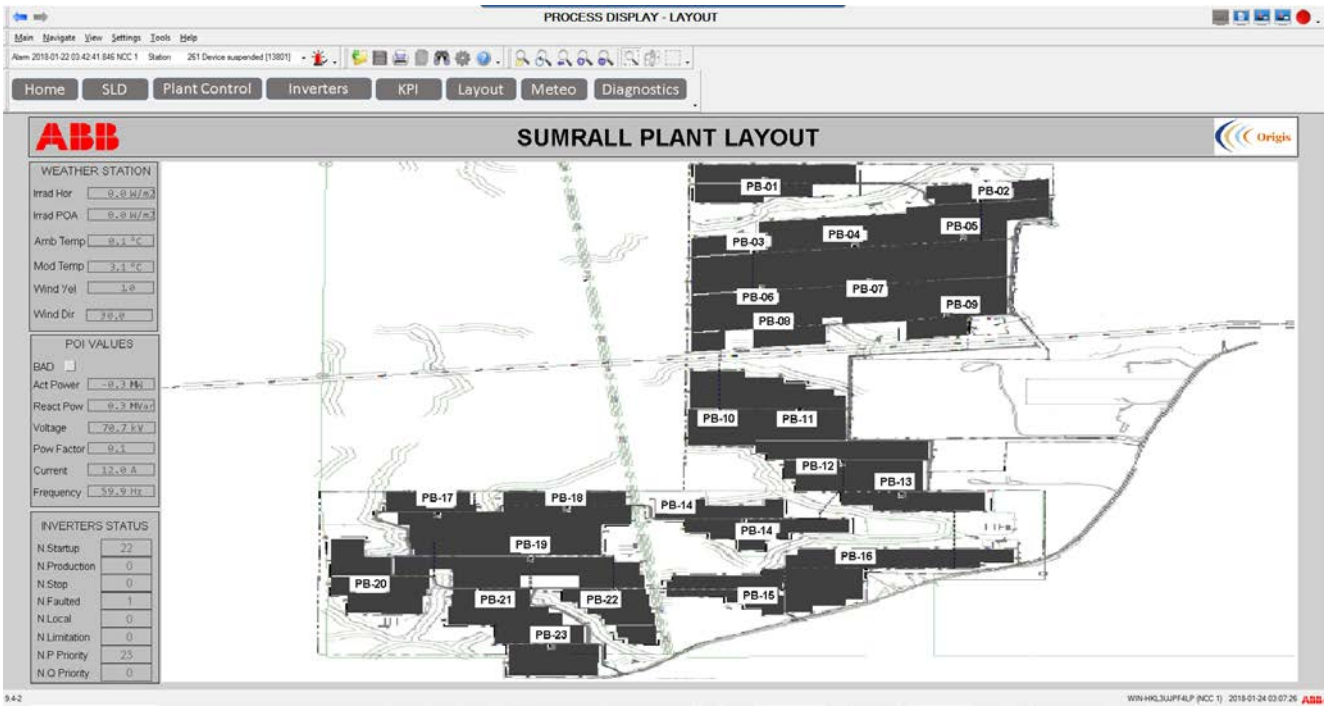

Figure 3-2 Home Page

If you close the Monitor PRO windows, the MicroSCADA service remains active and continues to perform data acquisition and elaboration.
To stop the MicroSCADA service you must click on the stop button in Control Panel and the data acquisition and processing will be terminated.

3.1 Login and Logout

In order to access the functions and utilities on MicroSCADA the user must login. A set of rights of the USER prevents the access to some functions (typically configuration functions) from not authorized users. As per default, MicroSCADA allows all the users to visualize the data, while it is asking for specific to access the operations of updating or change of the parameters of management of plant.

Each user is associated with a certain user profile defined by the system manager. The user must do the logon to get the rights to access the functions and the operations.

Start a Monitor Pro session by launching the SYS600 Monitor Pro program. The Login dialog is displayed when Monitor Pro is started

In the dialog, Select the Project application and type the username and the password, then click “Login”:

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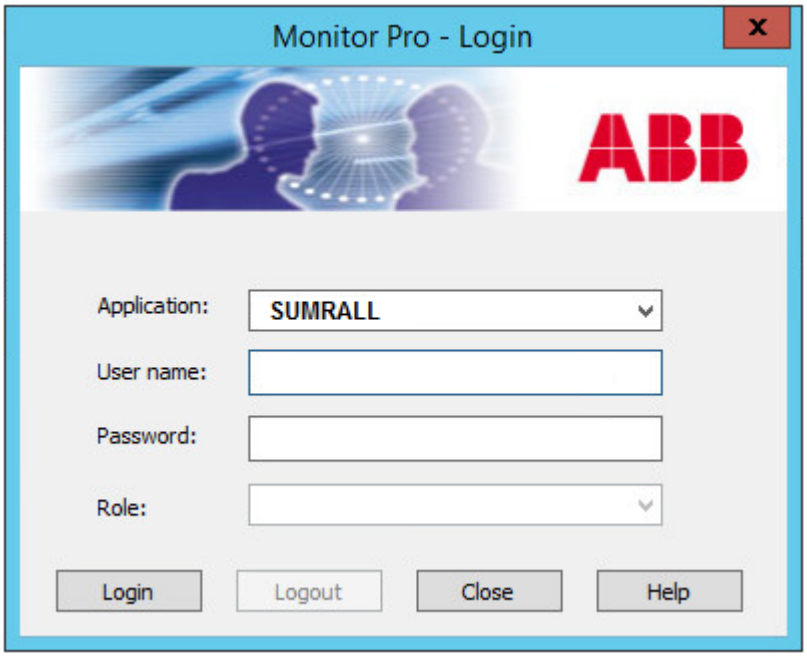

Figure 3-3 Monitor Pro Login

In Monitor Pro, logout means that the current user name and user authority are cleared. The options for logout are:

- Selecting **Main** > **Logout**.
- Closing the Monitor Pro by selecting **Main** > **Exit**.
- Automatic time based logout executed by Monitor Pro.

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4 HMI DESCRIPTION

4.1 Operator Station Interface

MicroSCADA Monitor PRO is the user interface to MicroSCADA and Plant Controller.
At start-up MicroSCADA Monitor PRO opens a workplace where user can see mimics, analyze alarms, open trend, etc.
Security is required to control access to MicroSCADA functions and utilities. A set of user rights prevents unauthorized users to access to particular functions (typically configuration functions).
Each user is identified with a user name, a password, a user role and authorization levels. A user must login, to obtain access rights to functions and operations.
Operations and actions in the system can be different for different roles. This assignment defines what authorization level a user needs to have in order to perform the operation or action. Examples of permissions are: Read (Viewer), Configure (Engineer) and Operate (Operator). This is defined using 'User Management Tool'.
Following chapters describe workplace, mimics and alarms configuration guidelines for Sumrall 52 MW PV Plant.

4.2 Workplace Organization

MicroSCADA Monitor Pro layout is divided in different areas as show in Figure 4-1

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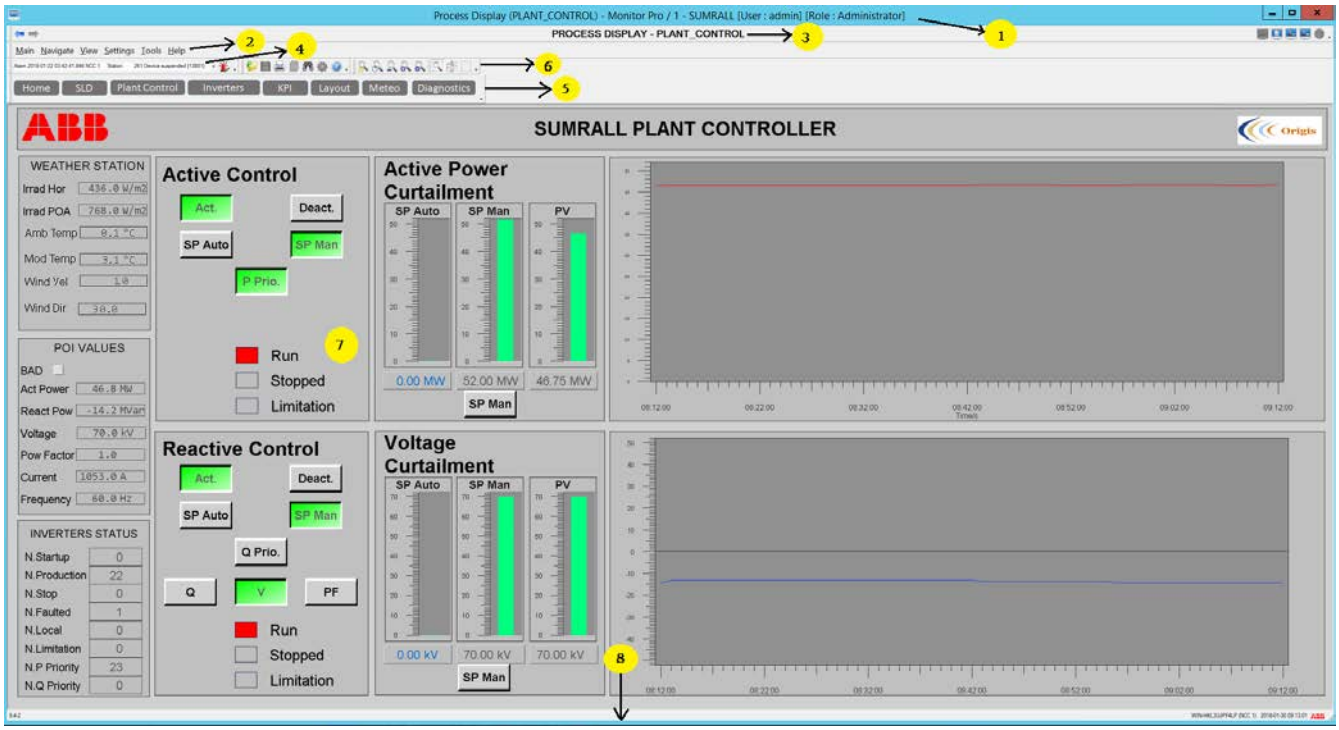


Figure 4-1 MicroSCADA pro work place

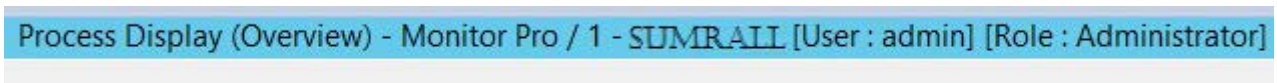
Default Monitor Pro layout contains the following components:

1. Process display name, application name, user name
2. Main menu bar
3. View Info toolbar
4. Alarm Row/Latest unacknowledged alarms
5. Process Display navigation shortcuts
6. Quick Access toolbars – Zoom/Pan/Print etc.
7. Process Display Area
8. Status bar

Monitor Pro Menus and Toolbars are explained in detail in the 'SYS600_Operation Manual' available along with SYS600 documentation

4.2.1 Process display name, application name, user name

This area shows the name of the displayed mimic, the application name which is currently used and the user who has logged in



4.2.2 Main Menu bar

Main Menu bar contains all the menus and tools which are available in MicroSCADA pro. The tools are accessible based on the user authorization and context.

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4.2.3 View Info tool bar

This toolbar displays the name of the current display, for e.g. the name of the process display/mimic or pre configuration name of Alarm List/Event Lists/Reports/Trends.

4.2.4 Alarm Row/Latest unacknowledged alarms

The Alarm row provides a quick notification of an alarming event in the system. The advantage is that it can be noticed easily, and it also instantly tells the operator what has happened and where. With the alarm row the alarms can easily be acknowledged. Display the alarm row by selecting it on the Toolbar tab

The alarm row shows all the unacknowledged active and inactive alarms in the system. The latest alarm is shown on the top of the list. Any of the alarms shown on the list can be selected to be acknowledged.

4.2.5 Process Display Navigation shortcuts

These are project specific display shortcuts to quickly access different mimic pages for e.g. SLD (Single Line Diagram), Meteo etc.

4.2.6 Quick Access toolbars

These toolbars provide the menu options inside the Main Menu bar in a quick access format. The options include Open, Print, Close Mimics as well as Zoom-in, Zoom-out functionalities.

4.2.7 Process Display Area

The Display Area is used to show operator windows including mimics, trends, alarm & event lists.

4.2.8 Status bar

The Status bar shows the SYS600 version number, Base system Node Name, (system name) and the current date and time.

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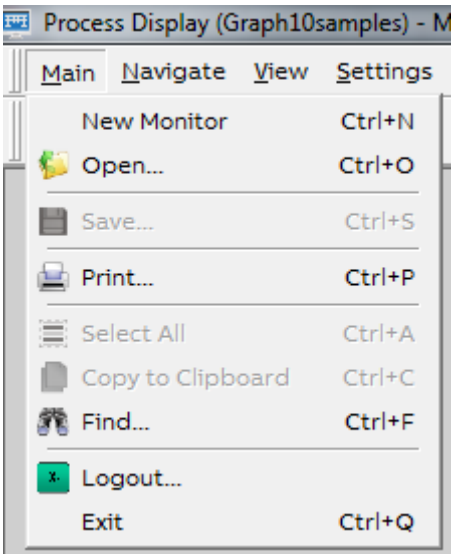
4.2.9 Main Menu bar

Main Menu bar contains all the menus and tools which are available in MicroSCADA pro. The tools are accessible based on the user authorization and context. Main tools inside the Menu bar are briefly mentioned below.

More details on toolbar functions can be found in MicroSCADA Standard documentation.

4.2.9.1 Main Menu



MENU OPTION	DESCRIPTION	Default User Role
New Monitor	New Monitor Pro Window	Administrator Engineer Operator Viewer
Open...	Open Process Display or Alarm List/Event List/Trend /Measurement Report Preconfigurations	Administrator Engineer Operator Viewer
Save...	Open the Save Preconfiguration	Administrator Engineer Operator* Viewer*
Print...	To navigate backward in last opened windows list	Administrator Engineer Operator Viewer
Select All	Select all visible values	Administrator Engineer Operator Viewer
Copy to Clipboard	Copies the selected visible trend/report data to the operating system clipboard.	Administrator Engineer Operator Viewer

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Find...	Find object in Process display or texts in Alarm/Event Lists	Administrator Engineer Operator Viewer
Logout...	Logout the current Monitor Pro session	Administrator Engineer Operator Viewer
Exit	Closes the Monitor Pro Window	Administrator Engineer Operator Viewer

*Limited functionality

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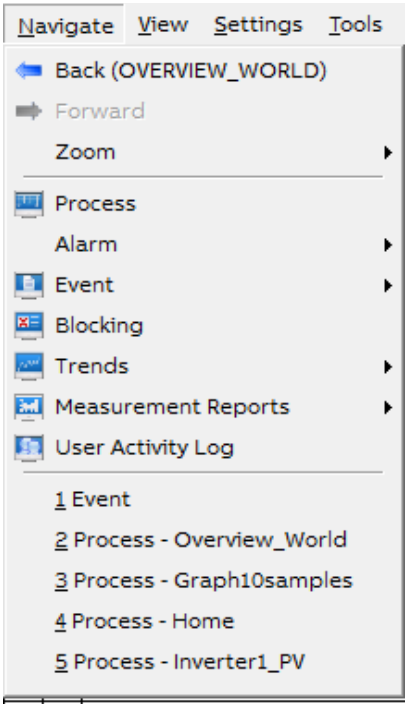
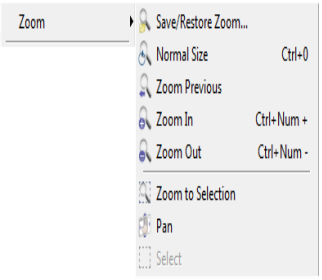
4.2.9.2 Navigate



MENU OPTION	DESCRIPTION	Default User Role
Back	Go back to previous display	Administrator Engineer Operator Viewer
Forward	Go forward to previous display	Administrator Engineer Operator Viewer
	Go to Zoom Display options. Works on Process display, Reports and Measurement Reports	Administrator Engineer Operator Viewer
Process	Navigate to Process display/Mimic	Administrator Engineer Operator Viewer
	Navigate to Alarm List (2 Templates available).	Administrator Engineer Operator Viewer

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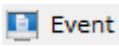
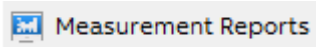
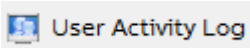
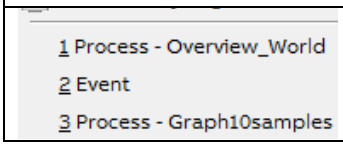
	Navigate to Event List	Administrator Engineer Operator Viewer
	Navigate to Blockin display	Administrator Engineer Operator Viewer
	Navigate to Trends display	Administrator Engineer Operator Viewer
	Navigate to Measurement Reports Page	Administrator Engineer Operator Viewer
	Navigate to the User Activity Log display	Administrator SECADM SECAUD
	Quick Navigate to recent displays	Administrator Engineer Operator Viewer

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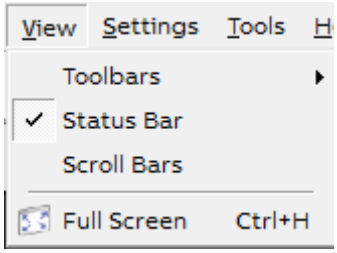
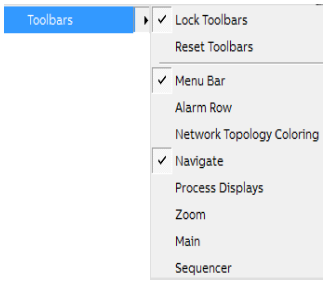
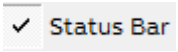
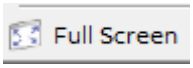
4.2.9.3 View



MENU OPTION	DESCRIPTION	Default User Role
	Option to show/hide different tool bars	Administrator Engineer Operator* Viewer*
	Show/Hide Status bar	Administrator Engineer Operator Viewer
	Show or hide scroll bars in process displays	Administrator Engineer Operator Viewer
	Make Monitor pro full screen.	Administrator Engineer Operator Viewer

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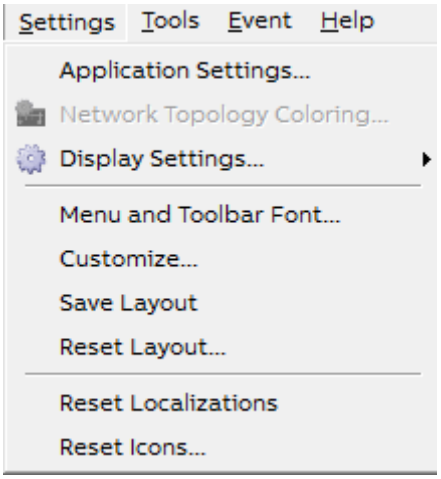
4.2.9.4 Settings



MENU OPTION	DESCRIPTION	Default User Role
Application Settings...	Open the application settings window	Administrator Engineer Operator* Viewer*
Network Topology Coloring...	Open Network topology coloring settings for electrical display(if applicable)	Administrator Engineer Operator*
Display Settings... / General Settings... / Layout Settings... / Color Settings...	Open Display specific settings for all displays except process display	Administrator Engineer Operator
Menu and Toolbar Font...	Change Menu and Tool bar font	Administrator Engineer Operator
Customize...	Create/Edit Toolbars, Menus/arrangement in Monitor Pro	Administrator Engineer Operator
Save Layout	Saves the current Monitor Pro lay out	Administrator Engineer Operator
Reset Layout...	Reset layout of toolbars and menu to saved/installation layout	Administrator Engineer Operator
Reset Localizations	Re-localizes Monitor Pro language	Administrator Engineer Operator
Reset Icons...	Reset Monitor Pro icons	Administrator Engineer Operator

*Limited functionality

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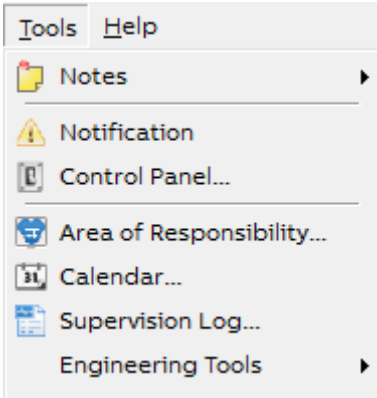
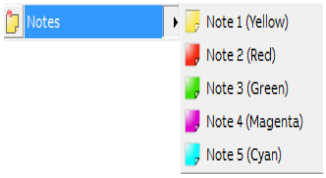
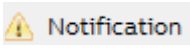
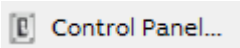
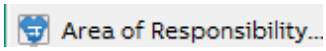
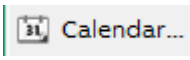
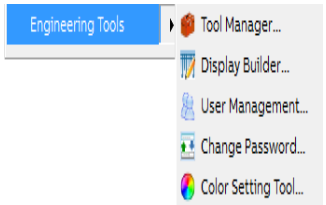
4.2.9.5 Tools



MENU OPTION	DESCRIPTION	Default User Role
	Quickly add Notes to Process displays	Administrator Operator
	Open Notify Window	Administrator Engineer Operator Viewer
	Open SYS600 Control Panel	Administrator Engineer
	Open area of responsibility window(if applicable)	Administrator Engineer Operator* Viewer*
	Open calendar tool(used for scheduling activities)	Administrator Engineer Operator
	Open System supervision log(if applicable)	Administrator Engineer Operator Viewer
	Open different Engineering tools.(Tool options are enabled only based on the authorization)	Administrator Engineer Operator* Viewer*

*Limited functionality

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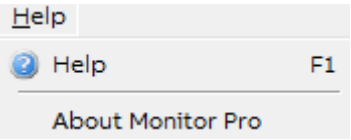
4.2.9.6 Help



MENU OPTION	DESCRIPTION	Default User Role
Help	Open SYS600 operation manual	Administrator Engineer Operator Viewer
About Monitor Pro	Show tool versions	Administrator Engineer Operator Viewer

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4.3 Mimics (Process Displays)

There are two families of mimics:

- Mimics for monitoring use
- Mimics for control use

Monitoring mimics can show real-time or averaged value from field. The averaged data are calculated from MicroSCADA calculation engine (time base: 15 minutes) to analyze performance.

At MicroSCADA Monitor PRO start-up the plant layout is showed.

Navigation is possible from mimics themselves with navigation button or sensible areas. On the top of all graphical pages, there is an area that permit the direct access to the graphical pages (buttons on the top left).

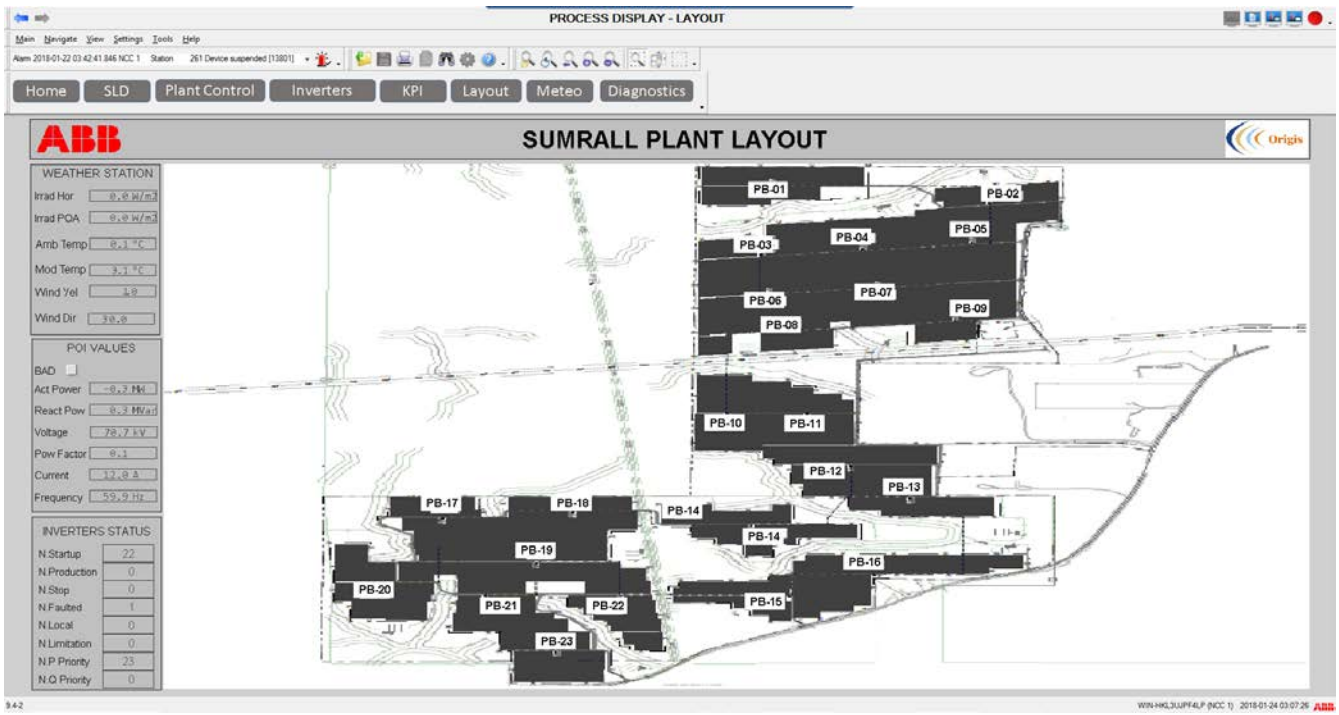

Figure 4-2 Sumrall Layout Page

This page shows the main plant data:

1. The status of the controls: if active power limitation and reactive power control are active or not, the status of the measure coming from the point of interconnection (POI), the status of the link with the Central Dispatch Center (external reference).
2. The POI data.
3. A summary of the Inverter status.

The plant layout displays the inverters and their status (a red rectangle indicate the inverter with errors). By clicking on the inverter the detailed page of that inverter is displayed.

There is another page that shows the main plant data (the Home Page):

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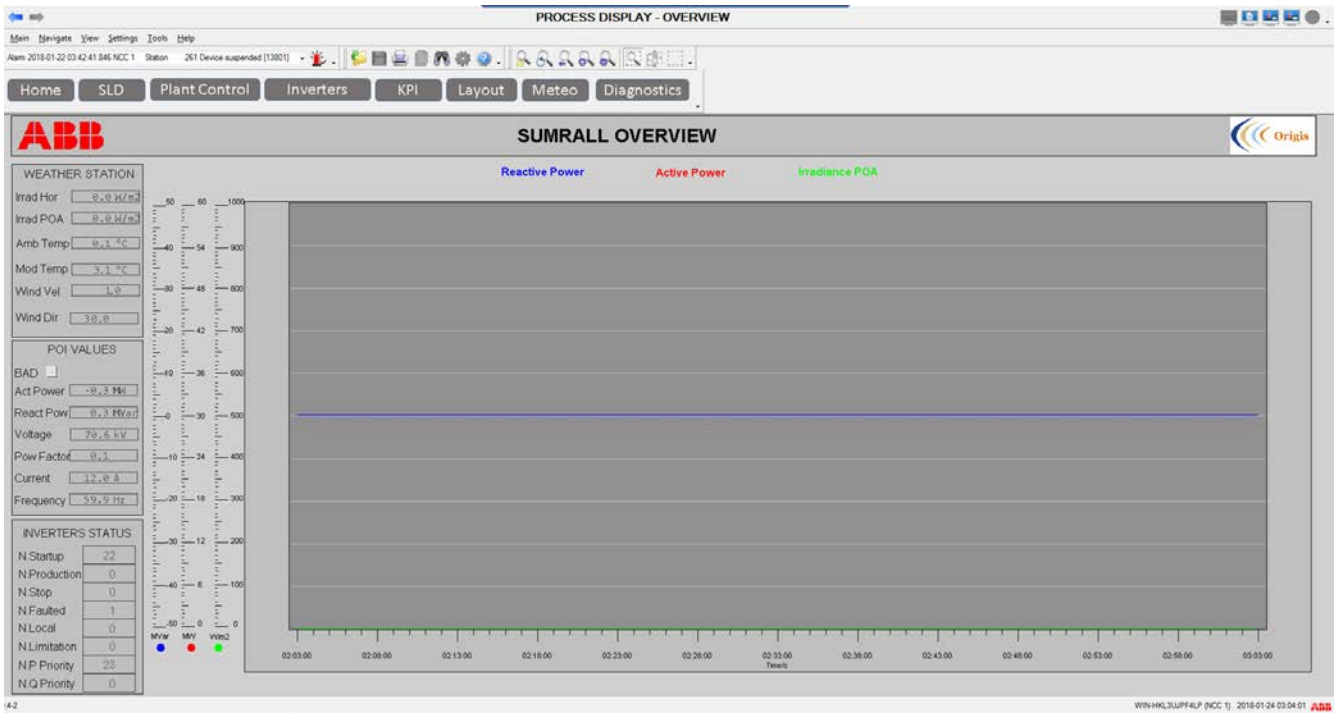

Figure 4-3 Sumrall Home Page

Like on the previous page there is an area on the left that shows:

1. The status of the controls: if active power limitation and reactive power control are active or not, the status of the measure coming from the point of interconnection (POI), the status of the link with the Central Dispatch Center (external reference).
2. The POI data.
3. A summary of the Inverter status.

Then the page contains a trend showing the last hours for the Active Power, Reactive Power, Irradiance on the Plane Of Array (POA).

Navigation buttons are located in the top left area, one click on the button open a new graphical page.

4.3.1 Plant Control Pages

There are two mimics dedicated to Plant Control.

Main control mimic displays (Figure 4-4) shows, starting from the left, all real time information about measures at POI, the status of the connection with the Central Dispatch Center (TSO), the statues for the inverters (Figure 4-5). Then there are the two area for the management of the Active Power limitation (Figure 4-6) and for the Reactive Power Control (Figure 4-8).

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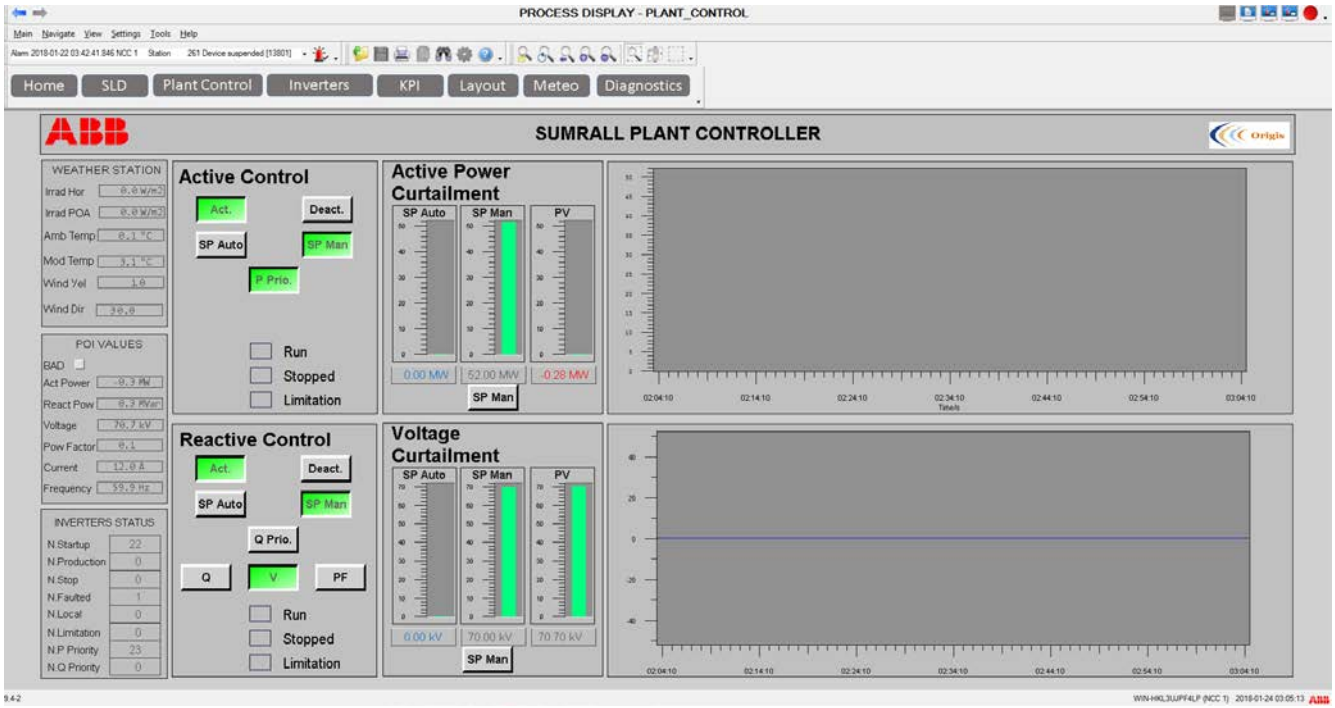

Figure 4-4 Main Plant Control HMI

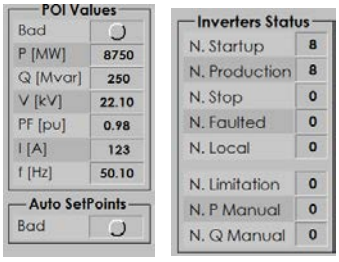

Figure 4-5 POI and Inverter Status

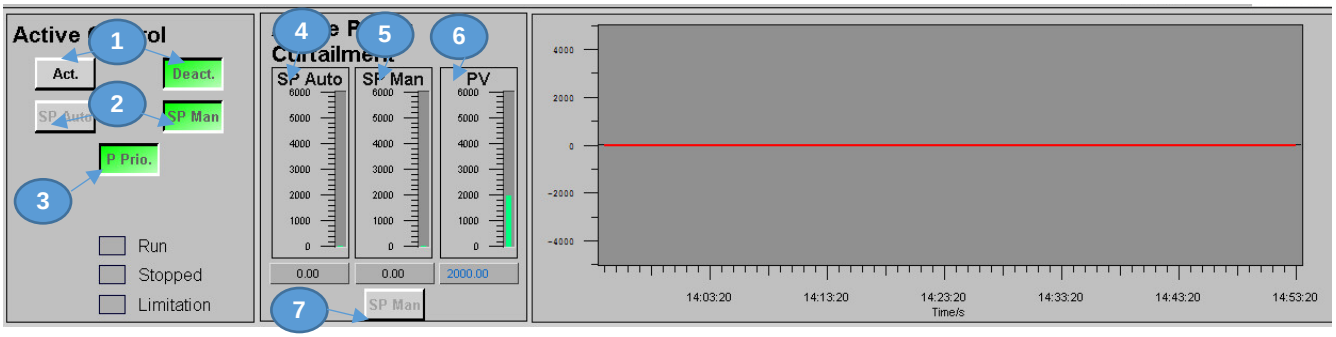

Figure 4-6 Active Power Limitation

Active Control

On the first box there are the buttons for activation and deactivation of the active power limitation (ref.1 on Figure 4-6). The green color will be applied to the button in order to show the actual status.

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Then other two button (ref.2) permits the selection of the mode: *Auto* means the set point arrive from the Central Dispatch Center; *Manual* means that the set point is inserted by the plant operator.

The button *P.Prio* (ref.3) indicates if the systems work in active power priority, that means the active power production is preferred to the reactive power, and vice versa.

Pressing the button *SP Man* (ref.7) will be opened a pop up that permits the operator to enter the set point for the Active Power Limitation (see Figure 4-7). The value can be increased or decreased with the buttons inside the pop up, or a precise value can be inserted clicking on the value and typing the requested value in the box.

The value for the set point coming from the Central Dispatch Center (rif.4), the value of the operator set point (rif.5) and the value of the process variable (rif.6) will be displayed.

Finally a trend with the last 30 minutes of data for the active power is displayed.

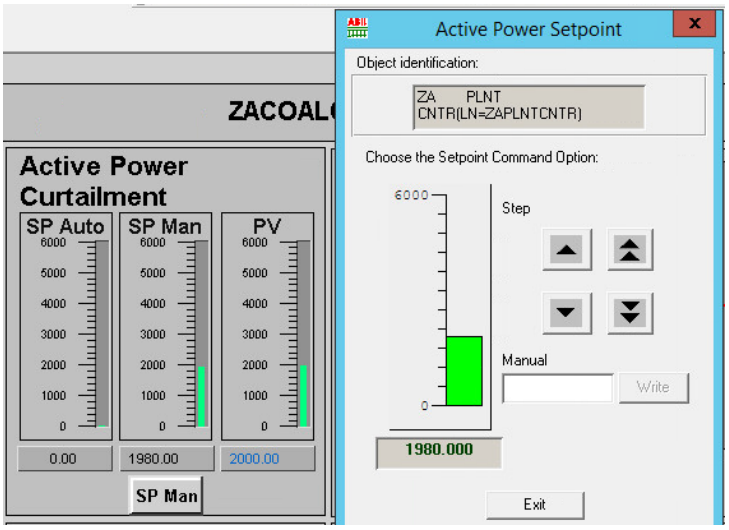

Figure 4-7 Active Power Limit Operator Set Point

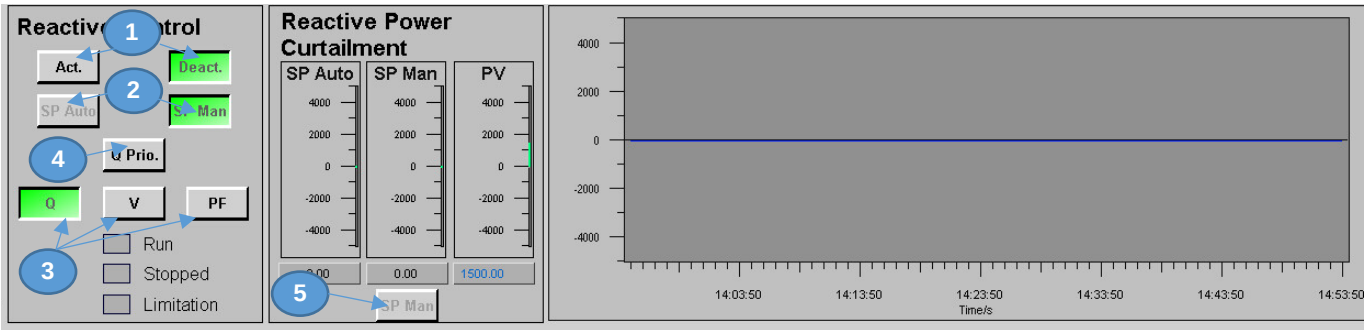

Figure 4-8 Reactive Power Control

Reactive Control

On the second box there are the buttons for activation ad deactivation of the reactive power (ref.1 on Figure 4-8). The green color will be applied to the button in order to show the actual status.

Then other two button (ref.2) permits the selection of the mode: *Auto* means the set point arrive from the Central Dispatch Center; *Manual* means that the set point is inserted by the plant operator.

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The operator will be able to select the control mode for the reactive power using the buttons indicated with *Q*, *V* and *PF* (ref.3). By means of these buttons he will select if the inserted set point is the value for the reactive power (Q), for the voltage (V) or for the power factor (PF).

The button *Q.Prio* (ref.4) indicates if the systems work in reactive power priority, that means the reactive power production is preferred to the active power, and vice versa.

Pressing the button *SP Man* (ref.5) will be opened a pop up that permits the operator to enter the set point for the Q, V or PF according to the selection (with buttons in ref 3) (see Figure 4-9). The value can be increased or decreased with the buttons inside the pop up, or a precise value can be inserted clicking on the value and typing the requested value in the box.

As for the active power control, the value for the set point coming from the Central Dispatch Center, the value of the operator set point, the value of the process variable and the trend with the last 30 minutes of data for the reactive power is displayed.

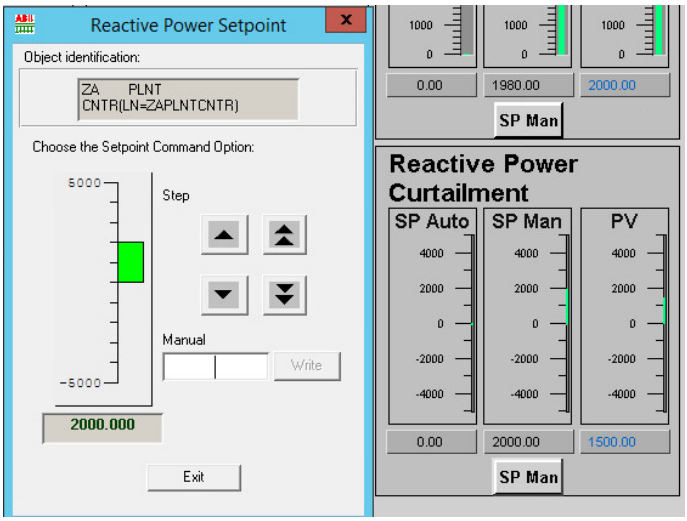

Figure 4-9 Reactive Power Operator Set Point

Clicking on the button Plant Control on the process display navigation shortcut, the operator will access to a menu (see Figure 4-10) with a way to access the pages with the control of the inverters under a certain feeder. The Feeder Control Page (Figure 4-11).

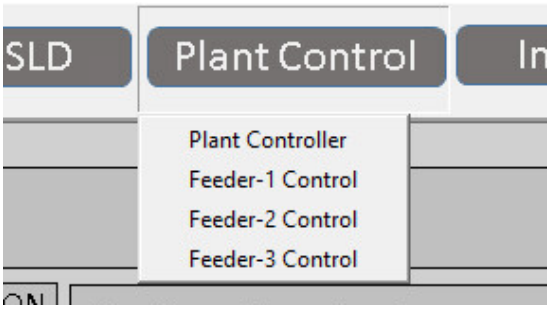

Figure 4-10 Selection of Inverter Control Page

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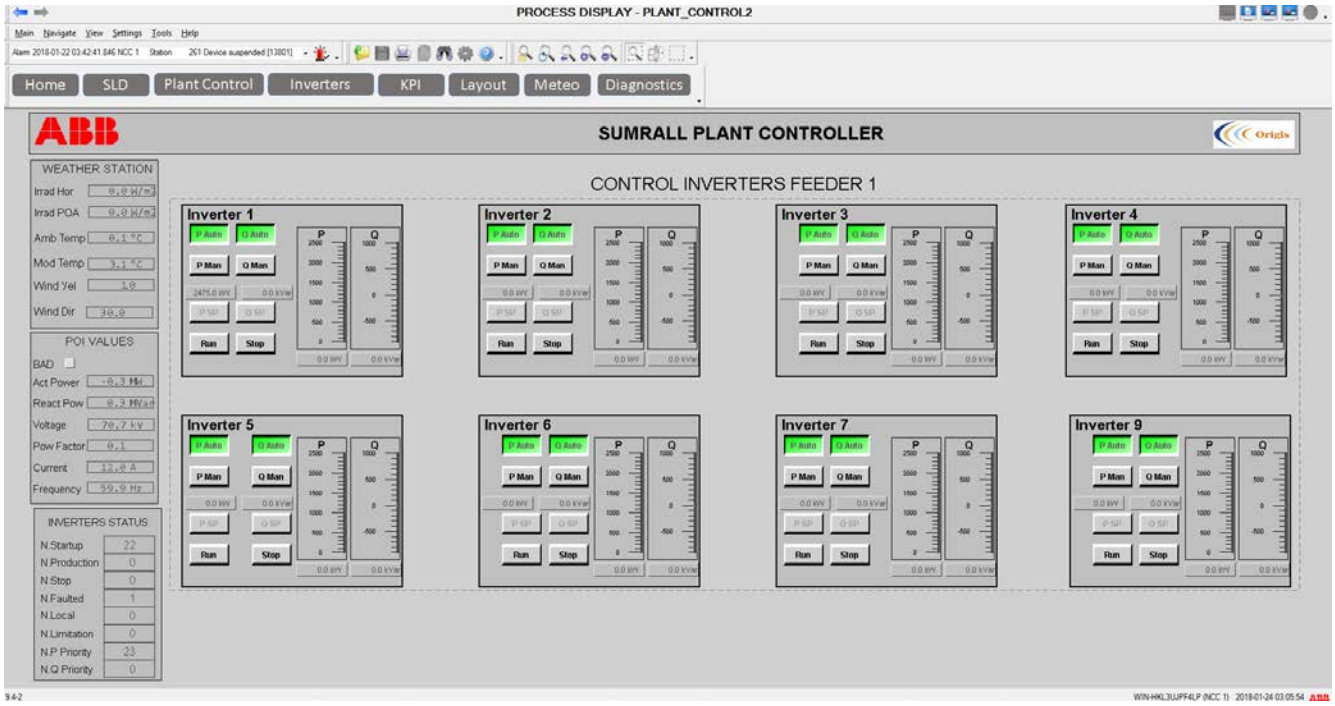

Figure 4-11 Feeder Control View

On this page the operator can operate on each inverter (of the selected feeder) individually: for each inverter he can send the start and stop command, put the inverter in manual mode and then send the set point of active and reactive power. From the Single line diagram the operator will be able to manage the start and stop of all the inverters and the status of automatic or manual for the active and reactive control (buttons on the bottom right of the SLD page).

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4.3.2 Inverter Station Monitoring Pages

On these pages the user has a complete and detailed view of production devices. On Figure 4-12 the page that shows the inverter1:

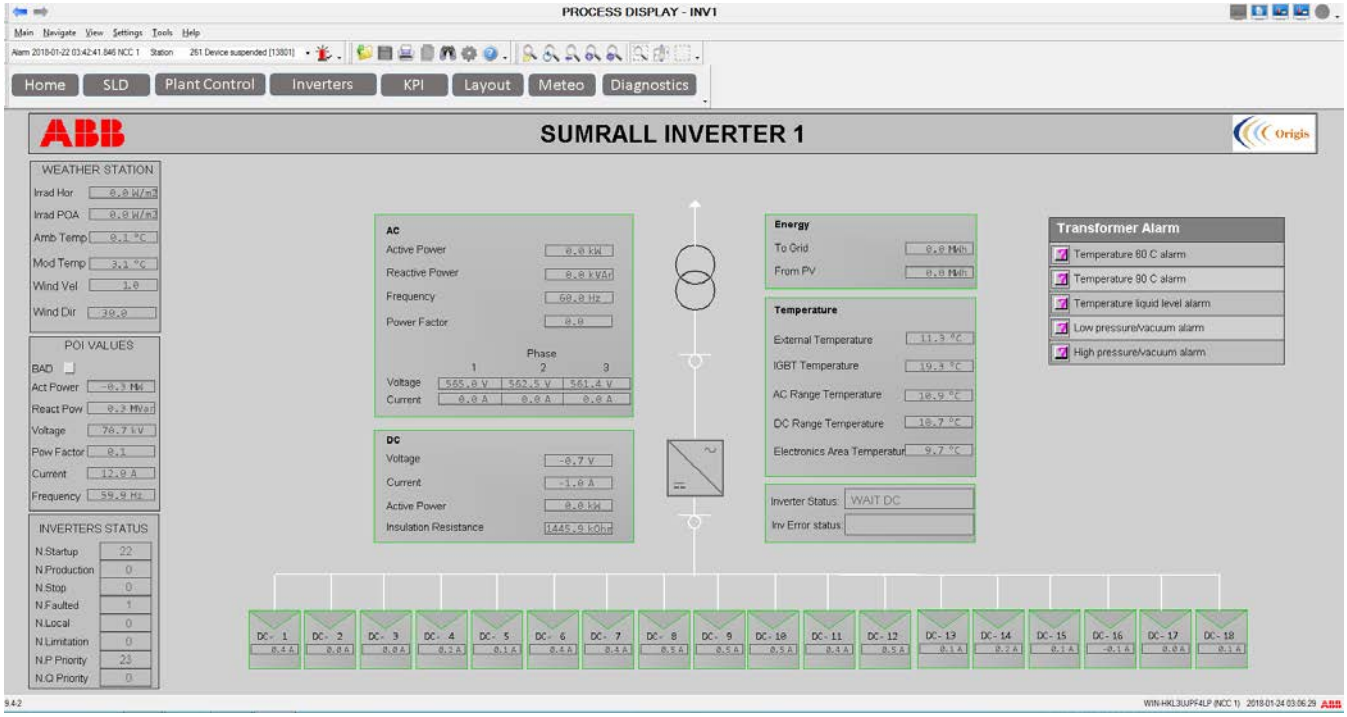

Figure 4-12 Inverter Station Monitoring Page

If inverter has an error status, a red exclamation mark will be presented close to the inverter error status. By clicking on this symbol, a page with the error description will be displayed.

By means of the button Inverter (on the process display shortcut area), see Figure 4-13, the user will be able to see the pages of others inverter stations.

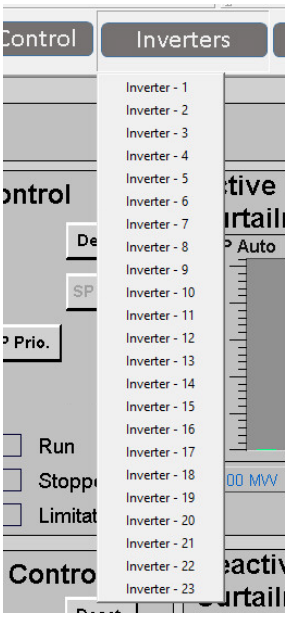

Figure 4-13 Inverter Station Monitoring Page Selection

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4.3.3 Communication Status Page

This page is a quick view on communication status.

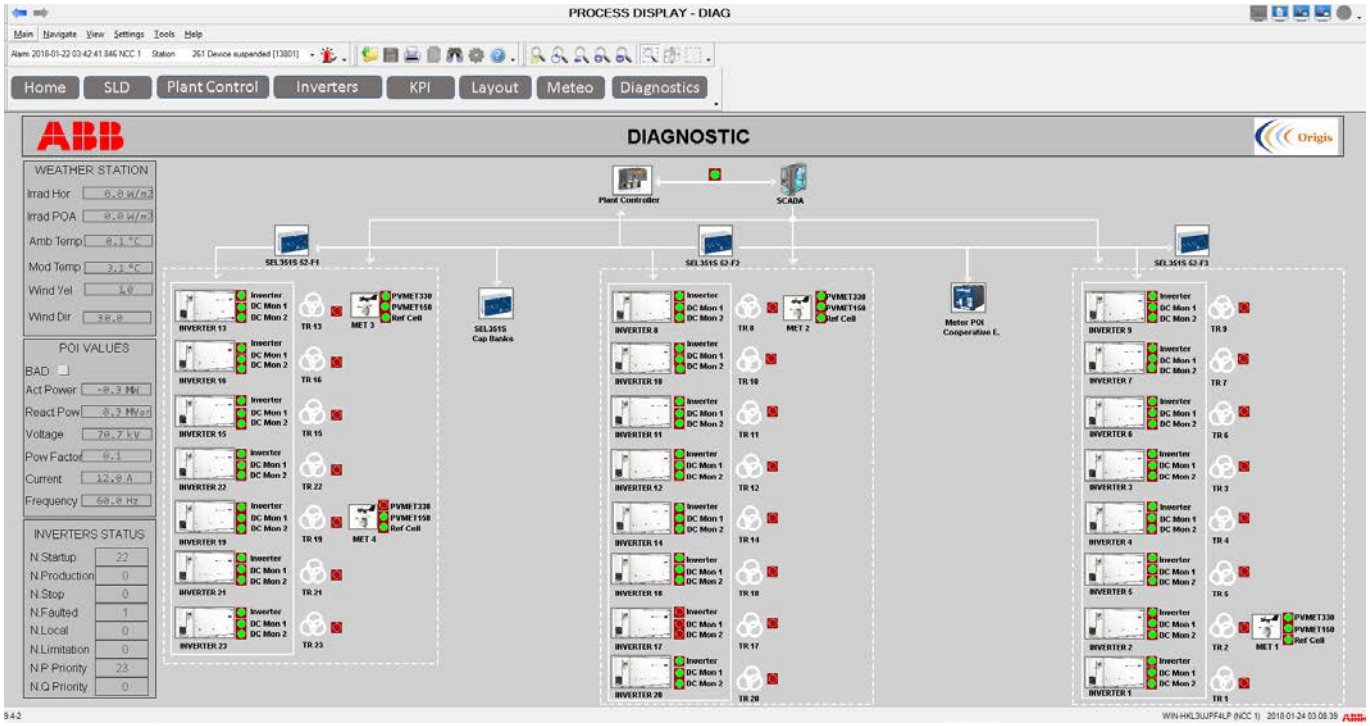

Figure 4-14 Communication Diagnostic Mimic

In this page the operator will see status of all the communication between MicroSCADA, AC500 Plant controller, Inverters, Metering at POI and Meteo Stations.

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4.3.4 Electrical View Pages

Electrical view (Figure 4-15) shows a simplified single line diagram of main buses and connected lines and it allows operator to monitor the main breakers, ground connector and measures.

In this page the operator finds the buttons for the command to close and open the breakers.

Table 4-1 lists dynamic behavior for each breakers type.

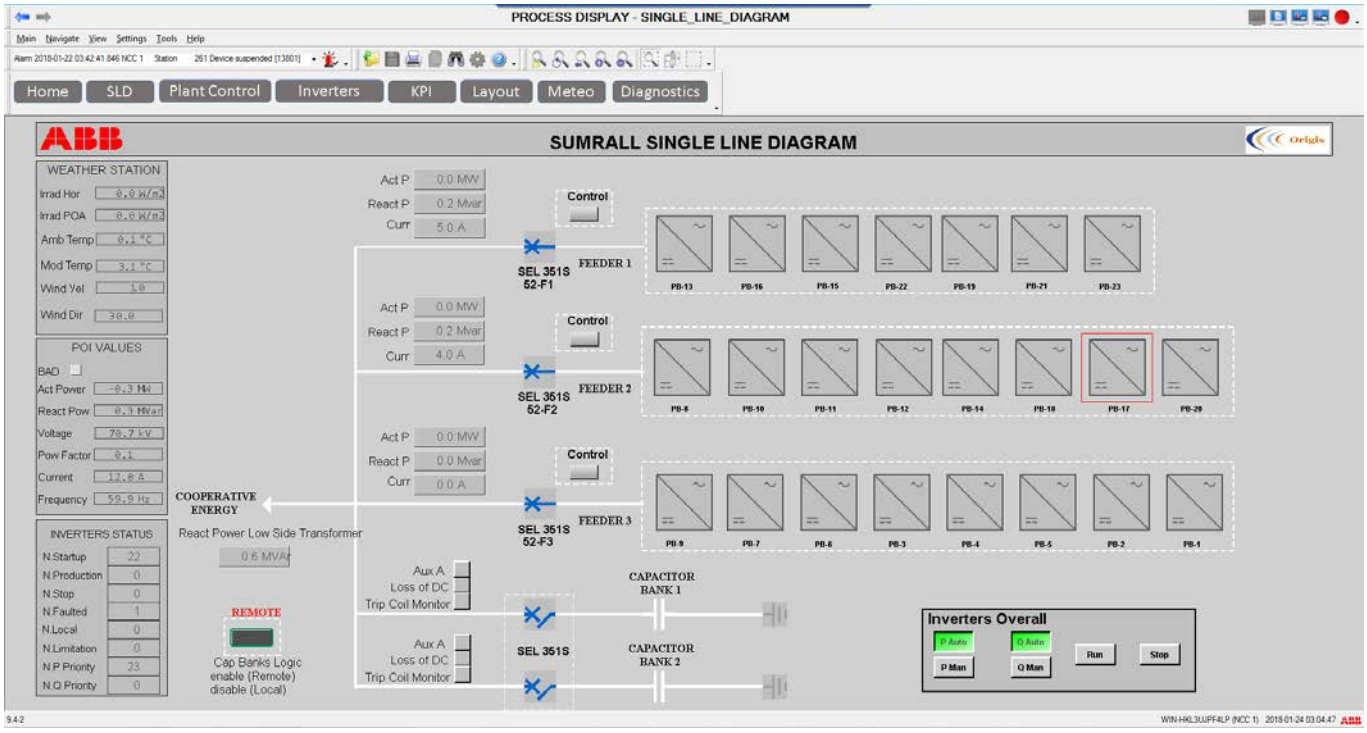


Figure 4-15 Electrical view

On the bottom right there are the buttons to put all the inverters in Auto or Manual mode and to Start or Stop all the Inverters (there is also a way to operate on each inverter, separately as described in The Feeder Control Page (Figure 4-11).

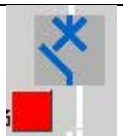
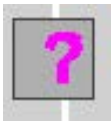
NAME	Closed	Opened	Intermediate	Trip	Bad Quality
Breaker					
Ground disconnector					

Table 4-1 Breaker dynamics

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On the bottom left there is the command to put the capacitor bank in Remote (that means the capacitor bank managed by the logic on the Plant controller) or Local (that means that the logic for the capacitor bank on the Plant controller, is disabled),

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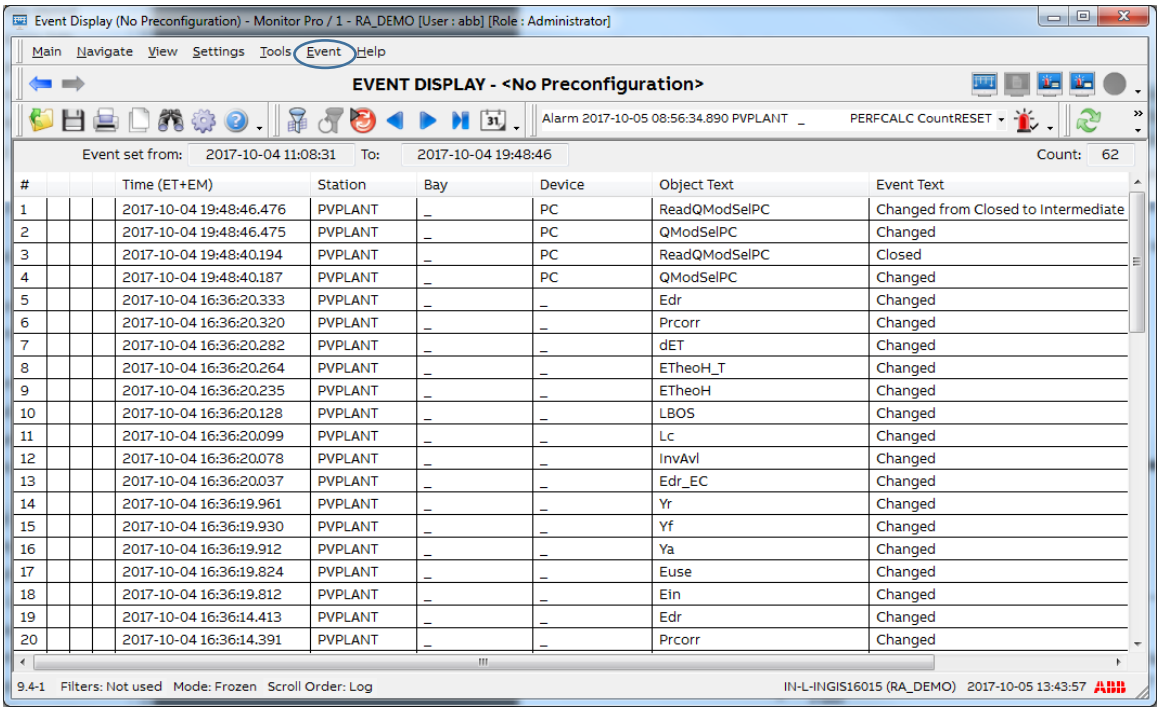
5 EVENT LIST

The Event List shows status changes of the various equipment in the plant. In addition, information about activities carried out by users, acknowledging of alarms, editing of limit values, and log on are listed. There are two ways to access the Event Display: selecting **Navigate** > **Event** or by clicking **Event Display** on the toolbar.

The Event Display presents the data in a structured way for the user's convenience. Each event is one row in the display. With default settings, Event Display rows consist of a time stamp, object identification, a signal text and a text indicating the status.



#	Time (ET+EM)	Station	Bay	Device	Object Text	Event Text
1	2017-10-04 19:48:46.476	PVPLANT	_	PC	ReadQModSelPC	Changed from Closed to Intermediate
2	2017-10-04 19:48:46.475	PVPLANT	_	PC	QModSelPC	Changed
3	2017-10-04 19:48:40.194	PVPLANT	_	PC	ReadQModSelPC	Closed
4	2017-10-04 19:48:40.187	PVPLANT	_	PC	QModSelPC	Changed
5	2017-10-04 16:36:20.333	PVPLANT	_	_	Edr	Changed
6	2017-10-04 16:36:20.320	PVPLANT	_	_	Prcorr	Changed
7	2017-10-04 16:36:20.282	PVPLANT	_	_	dET	Changed
8	2017-10-04 16:36:20.264	PVPLANT	_	_	ETheoH_T	Changed
9	2017-10-04 16:36:20.235	PVPLANT	_	_	ETheoH	Changed
10	2017-10-04 16:36:20.128	PVPLANT	_	_	LBOS	Changed
11	2017-10-04 16:36:20.099	PVPLANT	_	_	Lc	Changed
12	2017-10-04 16:36:20.078	PVPLANT	_	_	InvAvl	Changed
13	2017-10-04 16:36:20.037	PVPLANT	_	_	Edr_EC	Changed
14	2017-10-04 16:36:19.961	PVPLANT	_	_	Yr	Changed
15	2017-10-04 16:36:19.930	PVPLANT	_	_	Yf	Changed
16	2017-10-04 16:36:19.912	PVPLANT	_	_	Ya	Changed
17	2017-10-04 16:36:19.824	PVPLANT	_	_	Euse	Changed
18	2017-10-04 16:36:19.812	PVPLANT	_	_	Ein	Changed
19	2017-10-04 16:36:14.413	PVPLANT	_	_	Edr	Changed
20	2017-10-04 16:36:14.391	PVPLANT	_	_	Prcorr	Changed

Figure 5-1 Event Display Page

5.1 Event List Menu & Toolbar

The **Event** menu in the Main Menu Bar contains the following commands:

Filters: Opens a Filter Settings dialog, where filters can be selected and edited.

Reset Filter: Resets filters.

Comment: Opens a Comments dialog, which is used for writing comments to events.

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Keep Updating: Sets the Event Display to the updating mode. When changing to Event Display, the mode is by default set to updating mode, provided there is a user logged in.

Stop Updating: Sets the Event Display to the frozen mode. When changing to Event Display, the mode is by default set to frozen mode if no user is logged in.

Show Info Fields: Displays/hides the info fields.

Show Headers: Displays/hides the list headers.

Previous Event Set: Displays the list of previous events.

Next event set: Displays the list of next events.

Last Event: Displays the list of last events.

Select Day: Opens a Day select dialog.

Export: Exports the current view in CSV file format.

The toolbar is a shortcut that can be used in parallel with the drop-down menu.

The buttons in the toolbar from left to right are:

- Show Filters
- Reset Filter
- Switch to Updating or Frozen Mode
- Go to Previous Event Set
- Go to Next Event Set
- Go to Last Event
- Go to Selected Day

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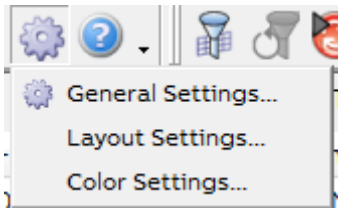
5.2 Event List Layout and Coloring

The display layout for e.g. the type of columns to be displayed as well as the coloring settings of the event row are configured using the display settings menu item.



Various conditions can be given for coloring a particular type of event

5.3 Features of Event List

- Scroll intervals are configurable from 10 to 100 pages, in steps of 10
- Fast navigation using the scroll bar: jump to the last page, one page back or forward.
- Possibility to a specific day
- Updating/Frozen presentation modes
- Manual input of additional comments for an event
- Possibility to display events either in log order or event order
- Printing of events on demand

If the Event List is in the updating mode, the list is updated for each new event. In the frozen mode (non-updating), a message is displayed to change the mode for the latest events. In the updating mode, a half a page of the list is displayed. New events are displayed in the lower half, until the page is full, and then is scrolled up a half page. When the Event List is in updating mode, the events are sorted in an order how they were received by the Field devices and written into history database, i.e. logging order.

To improve readability, event texts can be configured in a specific color.

Filters are used to display specific information. Filters can be modified and stored by the user.

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6 ALARM LIST

The Alarm Display shows a summary of the present alarm situation of the supervised process. Each alarm is normally presented as an alarm text row, which describes the cause of the alarm in the process. With default settings the alarm text row normally has a time stamp, an object identification, an object text and text indicating the alarm status.

See Figure 6-1 for Alarm Display Template 1 and Figure 6-2 for Alarm Display Template 2

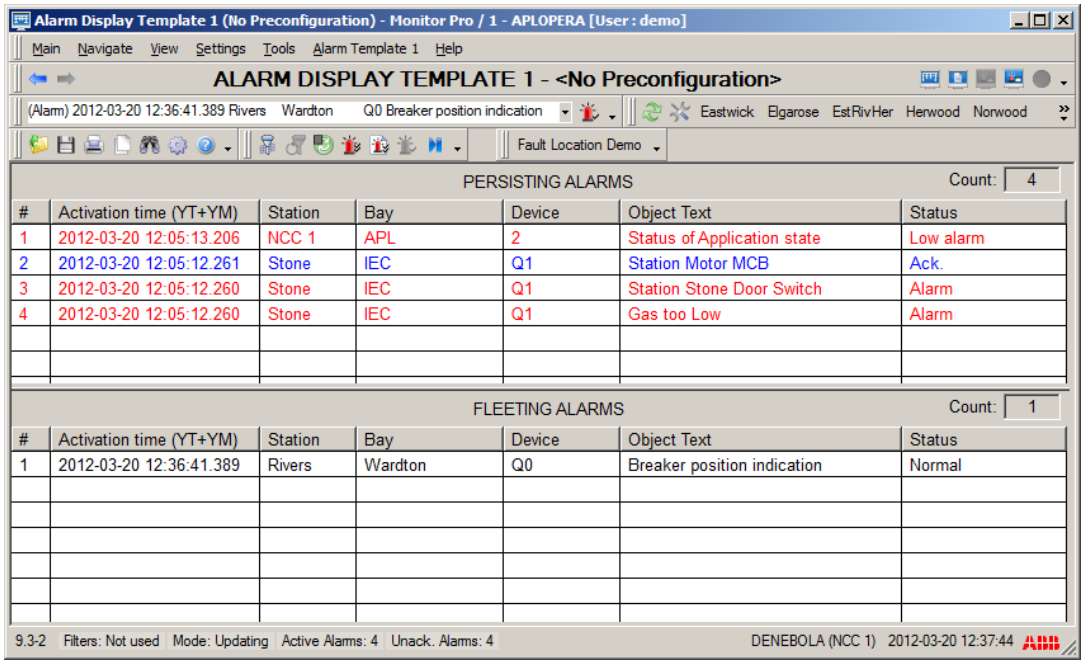

Figure 6-1 Alarm Display Template 1

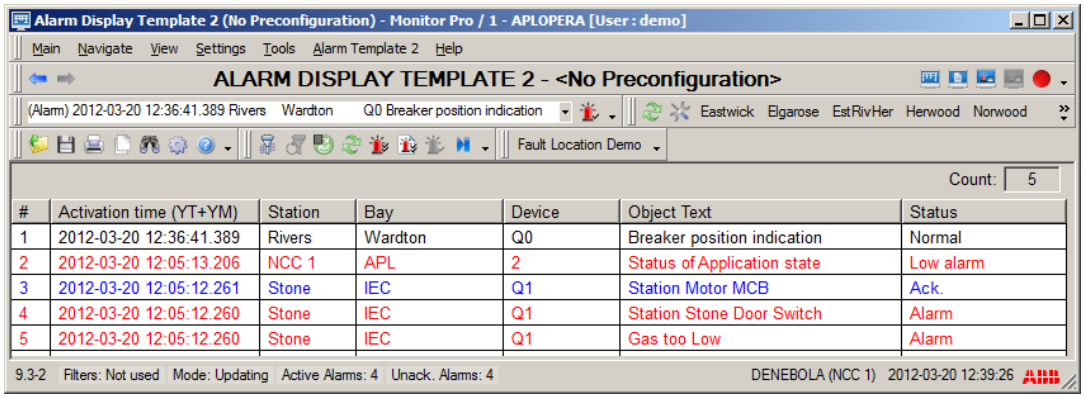

Figure 6-2 Alarm Display Template 2

In order to configure the alarms you should execute two steps:

Step 1 Alarms generation and alarm class definition and assignment of these to every process objects generating an alarm.

Step 2 Configuration, for each tag in the tag database, of the rules of the alarms (different for analog and digital tag type)

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6.1 Alarm List Menu and Tool Bar

The Alarm Display Template 1 and 2 menus contain the following commands:

Filters: Opens a dialog, where filters can be selected and edited.

Reset Filter: Resets filters back to the default settings.

Keep Updating: Sets the Alarm Display to the updating mode. When changing to Alarm Display, the mode is by default set to updating mode if a user is logged in.

Stop Updating: Sets the Alarm Display to the frozen mode. When changing to Alarm Display, the mode is by default set to frozen mode if no user is logged in.

Acknowledge…:

All: Acknowledges all the alarms on the list. A confirmation dialog is opened to confirm the operation.

Page: Acknowledges all the alarms on the current pages (both lists). A confirmationdialog is opened to confirm the operation.

Show Info Fields: Displays/hides the info fields.

Show Headers: Displays/hides the list headers.

Last Alarm: Scrolls the list to show the latest alarm and sets the mode to updating.

Export: Exports the current view in CSV file format.

The toolbar provides a shortcut to the commands in the menus. The toolbars in the Alarm Display Template 1 and Alarm Display Template 2 can be modified separately.

Figure 6-3 Toolbar of Alarm Display

The buttons in the toolbar of Alarm Display (Figure 6-3) are from left to right:

- Show Filters
- Reset Filter
- Switch to Updating or Frozen Mode
- Acknowledge All
- Acknowledge Page
- Acknowledge Selected Alarm
- Go to Last Alarm

The list can be set to two different modes: frozen and updating. When the list is in the frozen mode, it is not updated, and the alarm information can be read easily. If alarms are changed while the Alarm Display is in the frozen mode, the operator is notified with an informative text on the display area. When in the updating mode, the Alarm Display is updated when alarms are changed. The current mode is indicated in the Monitor Pro application window

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6.2 Alarm Rows

Each alarm is presented as a single alarm row. Different alarm templates list alarms in different states. The states are show in Figure 6-4.

With default settings, the different states are colored differently. The default colors are the background colors in the figure

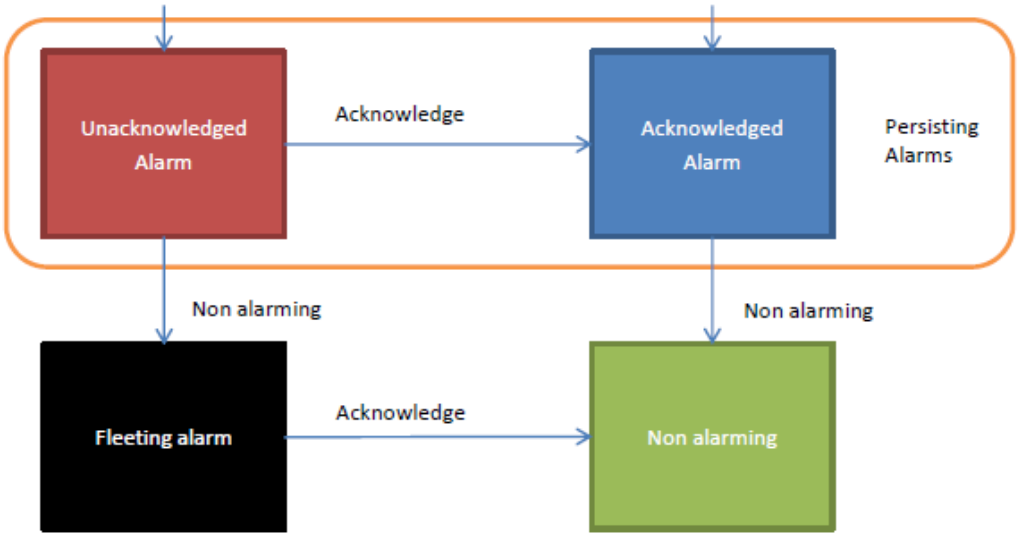
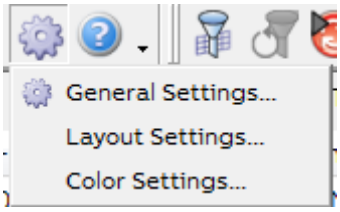


Figure 6-4 Alarm States

6.3 Alarm List Layout and Coloring

The alarm list layout for e.g. the type of columns to be displayed as well as the coloring settings of the alarm row are configured using the display settings menu item.



Various conditions can be given for coloring a particular type of alarm. Alarm class based coloring is also possible.

6.4 Alarm Classes

The alarms can be grouped into seven equally significant alarm classes

The alarm classes for this project are defined as shown in the list below:

- Urgent alarms (alarm class 1)
- Not urgent alarms (alarm class 2)
- Supervision alarms (alarm class 3)
- Common alarms (alarm class 4)
- Spare (alarm class 5)
- Spare (alarm class 6)

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- Spare (alarm class 7)

6.5 Alarm acknowledgment.

Alarm row

Acknowledge a single alarm by selecting the alarm from the dropdown list, see Figure 6-5.

Figure 6-5 Latest Alarms dialog

If the selected alarm is unacknowledged, the Acknowledgement dialog in Figure 6-6 can be opened by clicking the Acknowledge Selected Alarm button on the right-hand side of the Alarm Row.

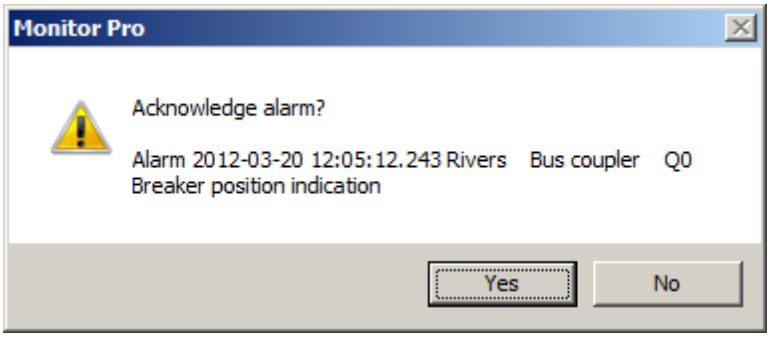

Figure 6-6 Acknowledge alarm dialog

In the dialog, the alarm text row (except for the status text) is shown to ensure that the right alarm is acknowledged. If Yes is clicked, the alarm will be acknowledged, the dialog closed, and the Alarm Display will be updated. Clicking No closes the dialog.

Alarm list

In the Alarm List, it is possible to **Acknowledge All** or **Acknowledge Page** from the corresponding toolbar or menu option. It is also possible to select the alarms to be acknowledged from the list by doing the following:

Press CTRL and click the alarms in the list to pick several alarms. Press SHIFT and click two rows on the list to select an area. When the alarms are selected, they can be acknowledged either from the context menu (right-click) or by using menu or toolbar option **Acknowledge Selected Alarm.**

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7 Color Setting Tool

Color Setting Tool is used to change the RGB values of logical colors. The tool contains tabs for status colors, topology colors and miscellaneous colors, which are related to the Process display. There are also dedicated tab sheets for the rest of the displays (Alarm / Event / Blocking / Trends / Measurement Reports display).

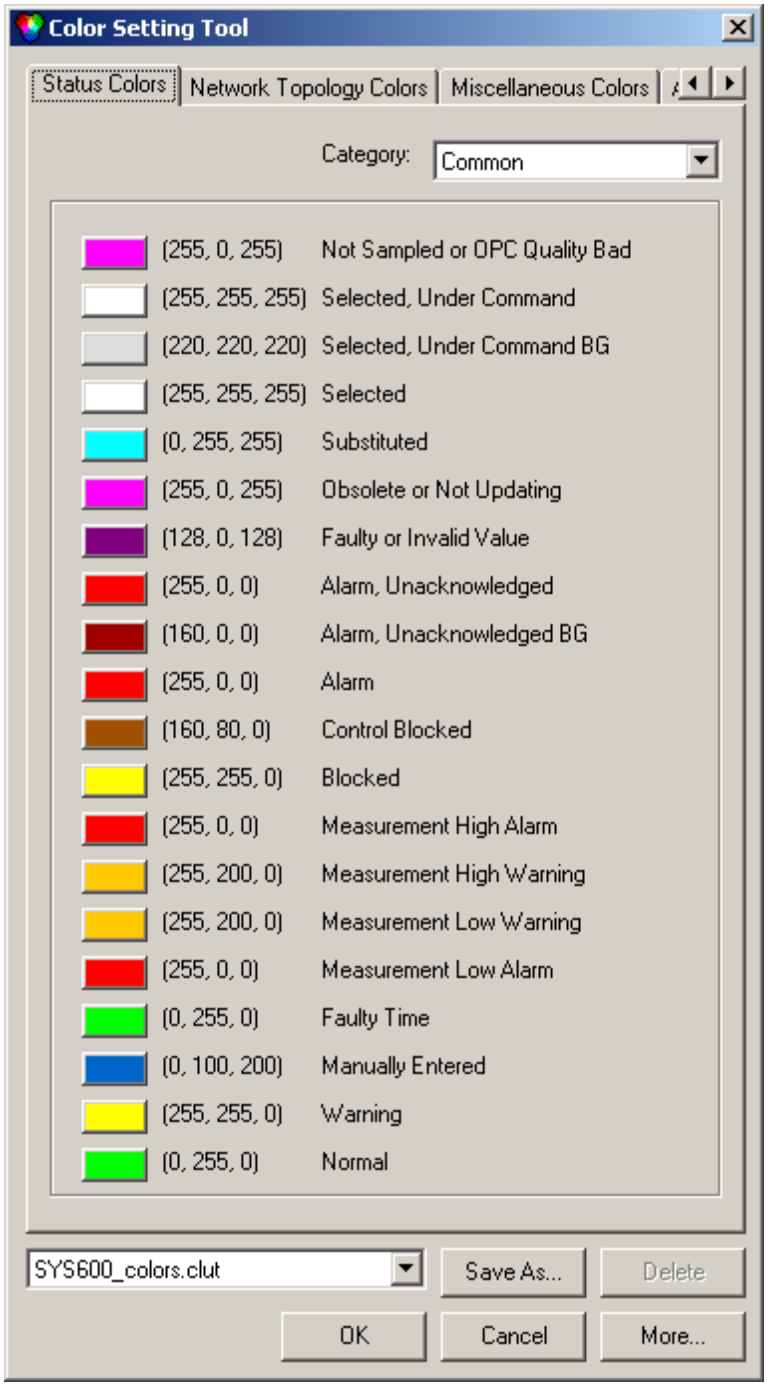

Figure 7-1 Color Setting Tool

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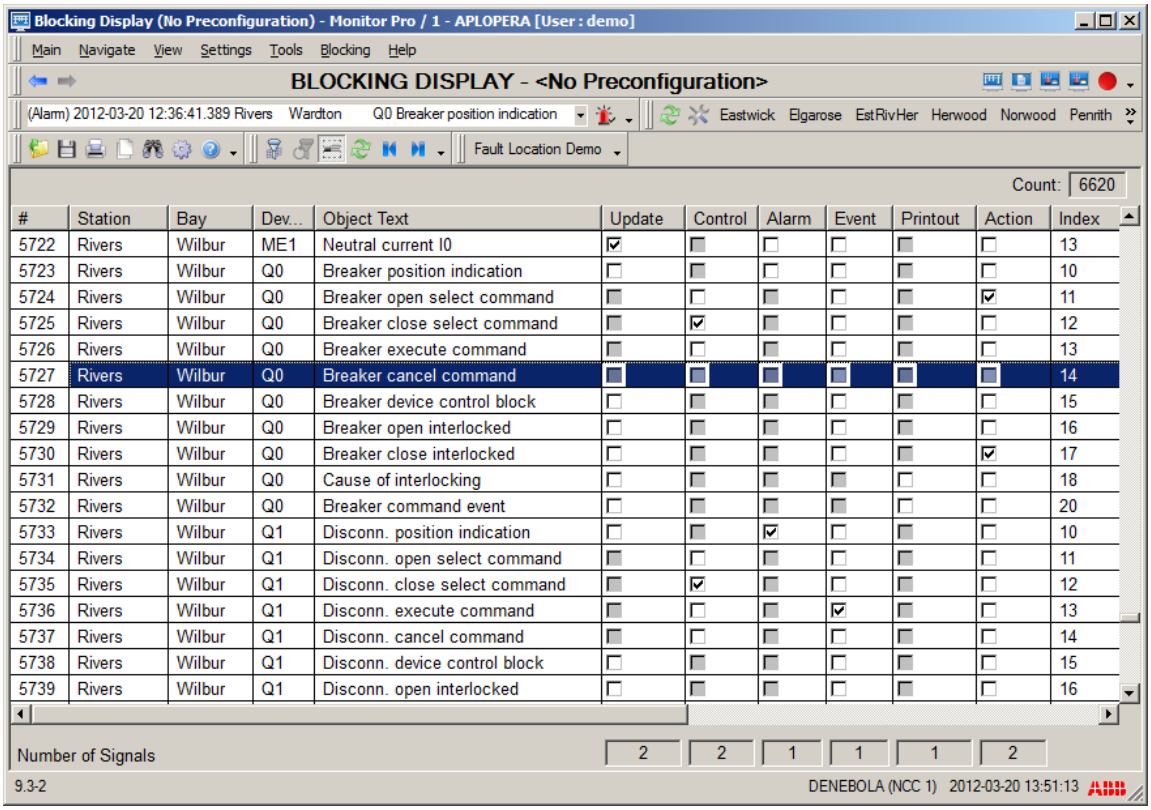
8 BLOCKING LIST

The Blocking Display summarizes the present blocking situation of signals in the database. Each signal is presented as a signal row, which describes the signal in the process. The signal text row normally consists of a signal text and a group of check boxes indicating the blocking state.



#	Station	Bay	Dev...	Object Text	Update	Control	Alarm	Event	Printout	Action	Index
5722	Rivers	Wilbur	ME1	Neutral current I0	☑						13
5723	Rivers	Wilbur	Q0	Breaker position indication							10
5724	Rivers	Wilbur	Q0	Breaker open select command						☑	11
5725	Rivers	Wilbur	Q0	Breaker close select command		☑					12
5726	Rivers	Wilbur	Q0	Breaker execute command							13
5727	Rivers	Wilbur	Q0	Breaker cancel command							14
5728	Rivers	Wilbur	Q0	Breaker device control block							15
5729	Rivers	Wilbur	Q0	Breaker open interlocked							16
5730	Rivers	Wilbur	Q0	Breaker close interlocked						☑	17
5731	Rivers	Wilbur	Q0	Cause of interlocking							18
5732	Rivers	Wilbur	Q0	Breaker command event							20
5733	Rivers	Wilbur	Q1	Disconn. position indication			☑				10
5734	Rivers	Wilbur	Q1	Disconn. open select command							11
5735	Rivers	Wilbur	Q1	Disconn. close select command		☑					12
5736	Rivers	Wilbur	Q1	Disconn. execute command				☑			13
5737	Rivers	Wilbur	Q1	Disconn. cancel command							14
5738	Rivers	Wilbur	Q1	Disconn. device control block							15
5739	Rivers	Wilbur	Q1	Disconn. open interlocked							16

Figure 8-1 Blocking Display Main View

The following blocking types are provided by the Blocking List:

- Alarm blocking
- Update blocking: indications are not updated by the process.
- Control blocking: operation commands are not sent to the process.
- Event blocking: event registrations are not made; events are not shown in the Event List.
- Printout blocking: events are not sent to the printer.
- Action blocking: event channel activation is blocked.

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9 TRENDS

Trend Application is used for trend analysis of measured values. The trend reports can be presented in graphical mode as full-graphic curves or in a tabular mode. These two modes share the same process data, but otherwise these modes can be used independently.

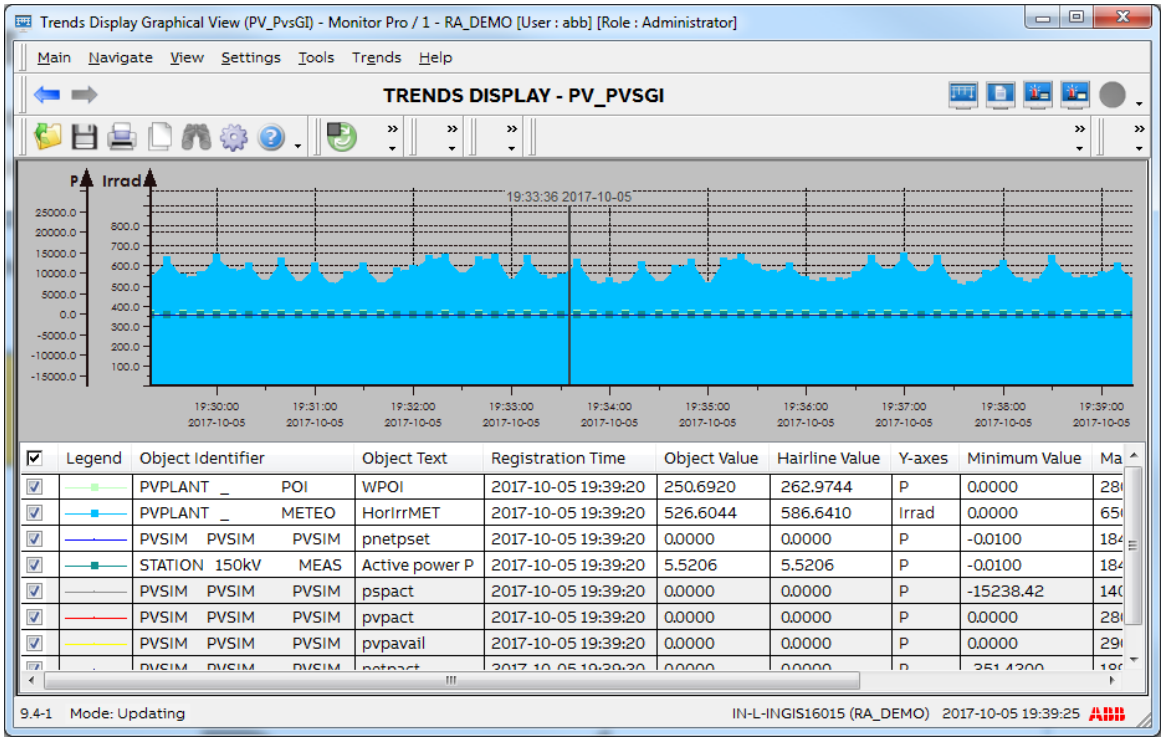

Figure 9-1 Trend Picture

Features of Trend application

- Graphic or tabular view modes
- Hairline function
- Zooming graphic view
- Scrolling with scroll bars and panning
- Configurable axes and line properties
- Using legend
- Update interval options from 30 seconds to 10 minutes
- Clearing trend data by user
- Save, Open and Delete Pre-configurations
- Printout option
- Update/Frozen modes

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10 REPORTS

Statistical measuring reports such as active and reactive power reports are provided with this functionality. Data for the reports are calculated and stored in real time. Report data are collected at periodic intervals. The most common method is to fetch raw data from the process, thereafter refine it and store it in the report database. Collection will be made within a time period of up to 2 years.

Each report columns constitute different types of data, e.g. points of time, measured, metered, manually entered or calculated values.

The reports can be presented in a tabular or in a graphical form. These two forms share the same report base.

General features:

- Daily report (active and reactive power)
- Monthly report (active and reactive power)
- Sum, Mean, Min and Max (in daily report)
- Sum (in monthly report)
- Selecting a specific time from the history
- Scrolling backward and forward in history.
- Scalable axis (in a graphical form)
- Scrolling in X and Y direction (in a graphical form)
- Load-duration curves (in a graphical form)
- Report data saving to a file, printout option
- Multiple report pages (Several report pages can be displayed within one report picture)

10.1 Starting Measurement Report Display

Start Measurement Reports by selecting ***Navigate > Measurement Reports*** *and select the appropriate report type (see Figure 10-1)*

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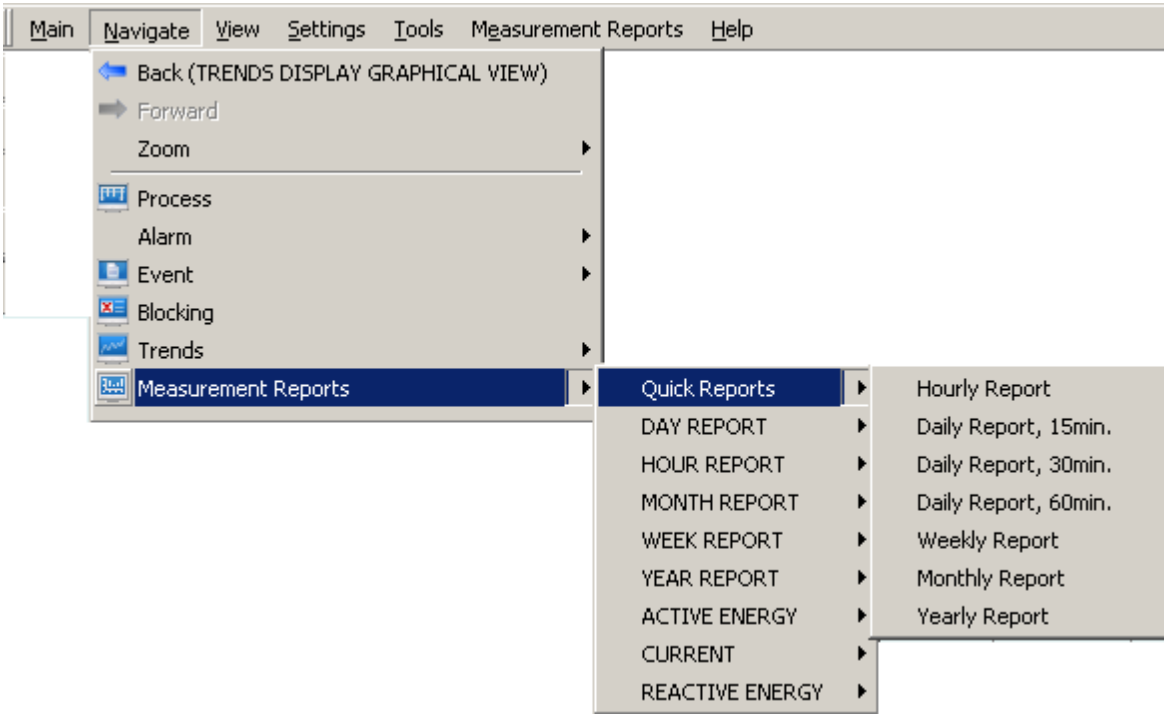

Figure 10-1 Opening Measurement Report Display

10.2 Measurement Reports Display Toolbars

Measurement Reports has four toolbars- Main, Display, Navigate and Zoom. Three toolbars are used for both views and Zoom toolbar is dedicated for the graphical view.

Figure 10-2 Main Toolbar

The Measurement Reports Main toolbar buttons from left to right are as follows:

- Open Preconfiguration
- Save Preconfiguration
- Print Display
- Copy to Clipboard from Selected Display
- Find in Selected Display
- Show Display Settings
- Show Help

Figure 10-3 Display toolbar

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The Measurement Reports display toolbar buttons and drop-down lists from left to right are as follows:

- Switch updating/frozen mode
- Refresh
- Show/hide report data
- Switch tabular/graphical view
- Select report page
- Select time interval

Figure 10-4 Navigation Toolbar

The Measurement report Navigation toolbar buttons from left to right are as follows:

- Go to first available period
- Go to previous period
- Go to next period
- Go to last available period
- Select period

10.3 Report Display Header

The header contains the following information fields from left to right:

- Period field
- Measurement Unit information

Figure 10-5 Measurement Reports header

The Period field gives information about the date/time range for the current visible data. The field content depends on the current active report type:

- For hourly and daily reports the current active day, as shown in Figure 10-5.
- For weekly reports the current active week.

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- For monthly reports the current active month.

  Period: October 2012 (W 40 - W 44)

- For yearly reports the current active year.
- Period: Year 2012

10.4 Report Type

In Sumrall application, there is only the daily report.

Report type	Time Column
Daily	Hour of the day and minute (hh:mm) step 5 minutes

10.4.1 Daily Report Plant: Production and Performance

Following an example:

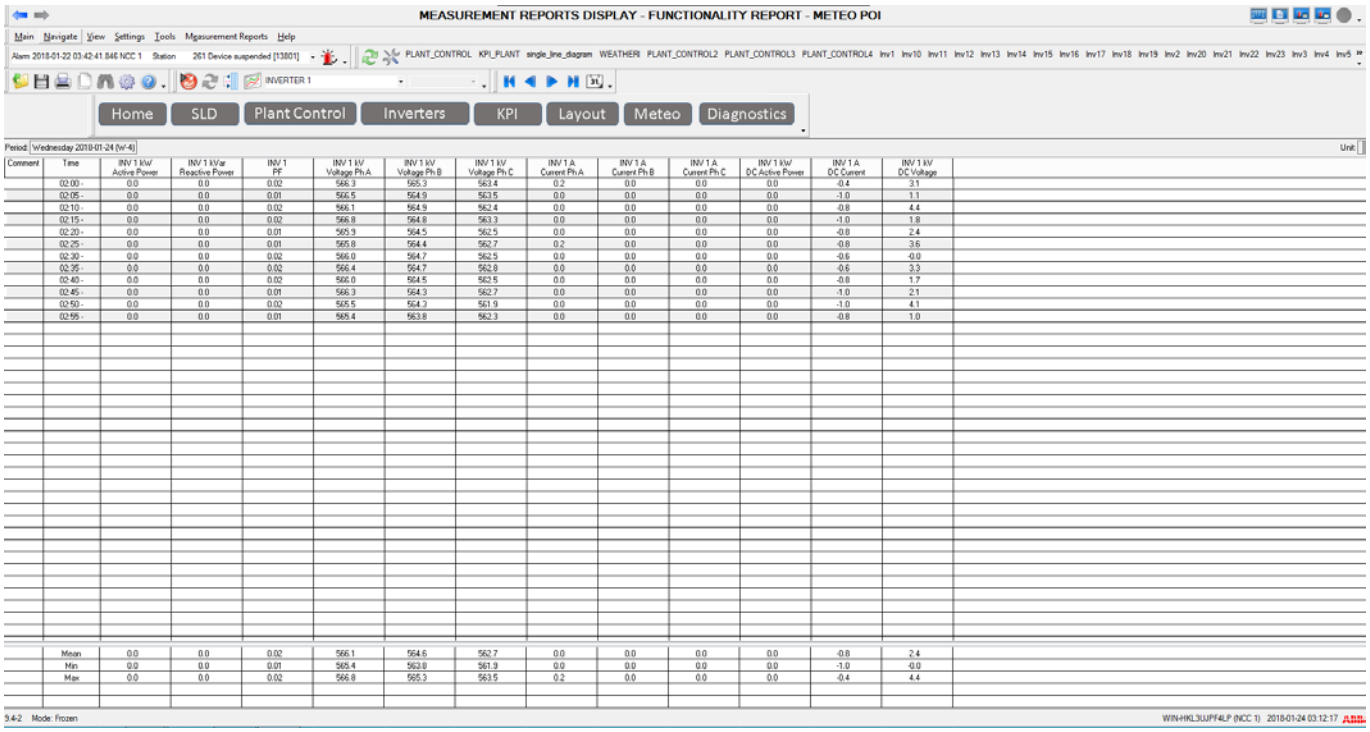

Figure 10-6 Daily Plant Report Example

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10.5 Export Reports

The selected report page can be manually saved to a file in .CSV format. It is possible to save the selected Report data to a file in .CSV format. In .CSV format, the separator between the columns is retrieved from the system settings. It can be changed in Windows control panel > Region and Language > Additional settings... > Customize Format numbers tab > List separator.

1. Select Export from the Measurement Reports menu. The Save As dialog is opened.
2. Specify the folder and the file name.
3. Click Save to export the data.

The exported text file contains the header information, the report time period, the unit information and the report data. When opening the export file with Microsoft Excel, select **Format** > **Cells** > **Text** (in the Category list).
This way the time format will be displayed correctly.

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11 MAINTENANCE

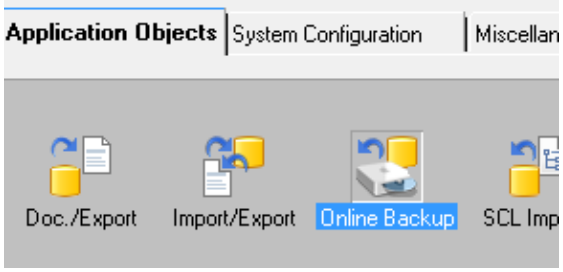
11.1 BACK-UP

With Backup Tool the user can make an online backup of the MicroSCADA application. MicroSCADA is running while the backup is made.

This tool makes a backup of the main application, not of the whole MicroSCADA. In other words, it makes a backup of everything located under *\SC\APL\'main application name'* This includes the whole database, calculations, reports, Toolbars, customizations etc.

This means that a second backup of *\SC* is always needed if *SYS_BASCON.COM* or other files located outside the application are changed

Backup tool is started from Tools→Tool Manager→Application Objects→Online Backup



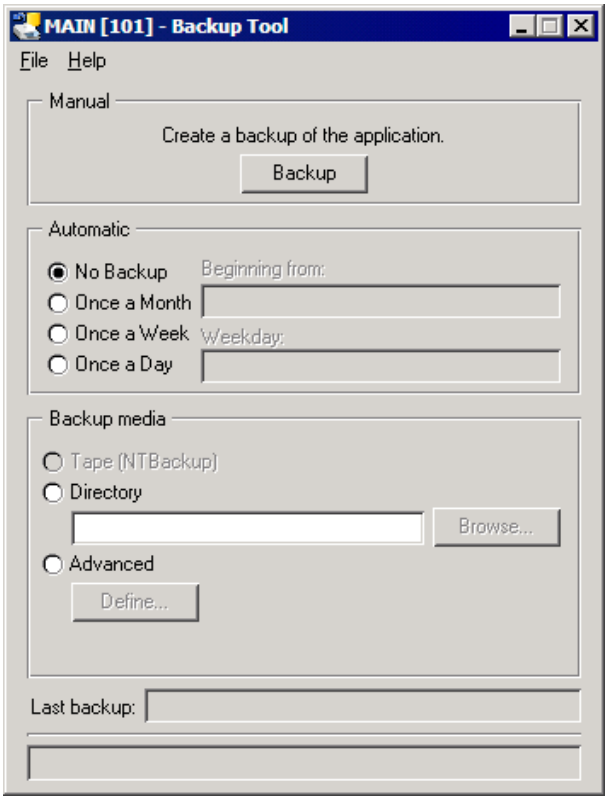


Figure 11-1 Backup Tool

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Manual Backup

Click Backup in the Backup Tool window shown in Figure 11-1, and after that click Yes. The procedure may take several minutes or even hours depending on the size of the application.

When a manual backup is started from this tool, it will trigger the command procedure BCK_START, which starts the backup.

Auto Backup

When No Backup is selected in the Backup Tool window shown in Figure 11-1, no automatic backup is made. If once a month is selected, the time channel BACKUP_1 activates the start of backup.

If once a week is selected, the BACKUP_2 activates the start of backup, and if Once a Day is selected then BACKUP_3 activates the start of backup.

As default, the time channels are started as shown in Table 11-1. If there is a need to change the time, it can be done with the Time Channel Tool, which is opened from the application object navigator.

BACKUP_1 First day every month at 00:10 (once a month)	BACKUP_1 First day every month at 00:10 (once a month)
BACKUP_2 Every Friday at 18:01 (once a week)	BACKUP_2 Every Friday at 18:01 (once a week)
BACKUP_3 Every day at 00:10 (once a day)	BACKUP_3 Every day at 00:10 (once a day)

Table 11-1 Backup Time Channels

More details about the backup are available in the ‘SYS600 Installation and Administration Manual’

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